Operations Research

Network flow problems

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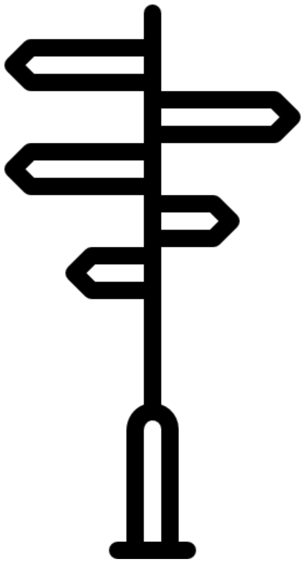
Course outline



- **Linear Optimization**
  - Modeling by linear programming and graphical solution
  - Solving linear programs, convexity
  - Simplex algorithm, 2-phase simplex
  - Simplex algorithm in matrix notation
  - Sensitivity analysis, re-optimization
  - Duality theory, zero-sum games

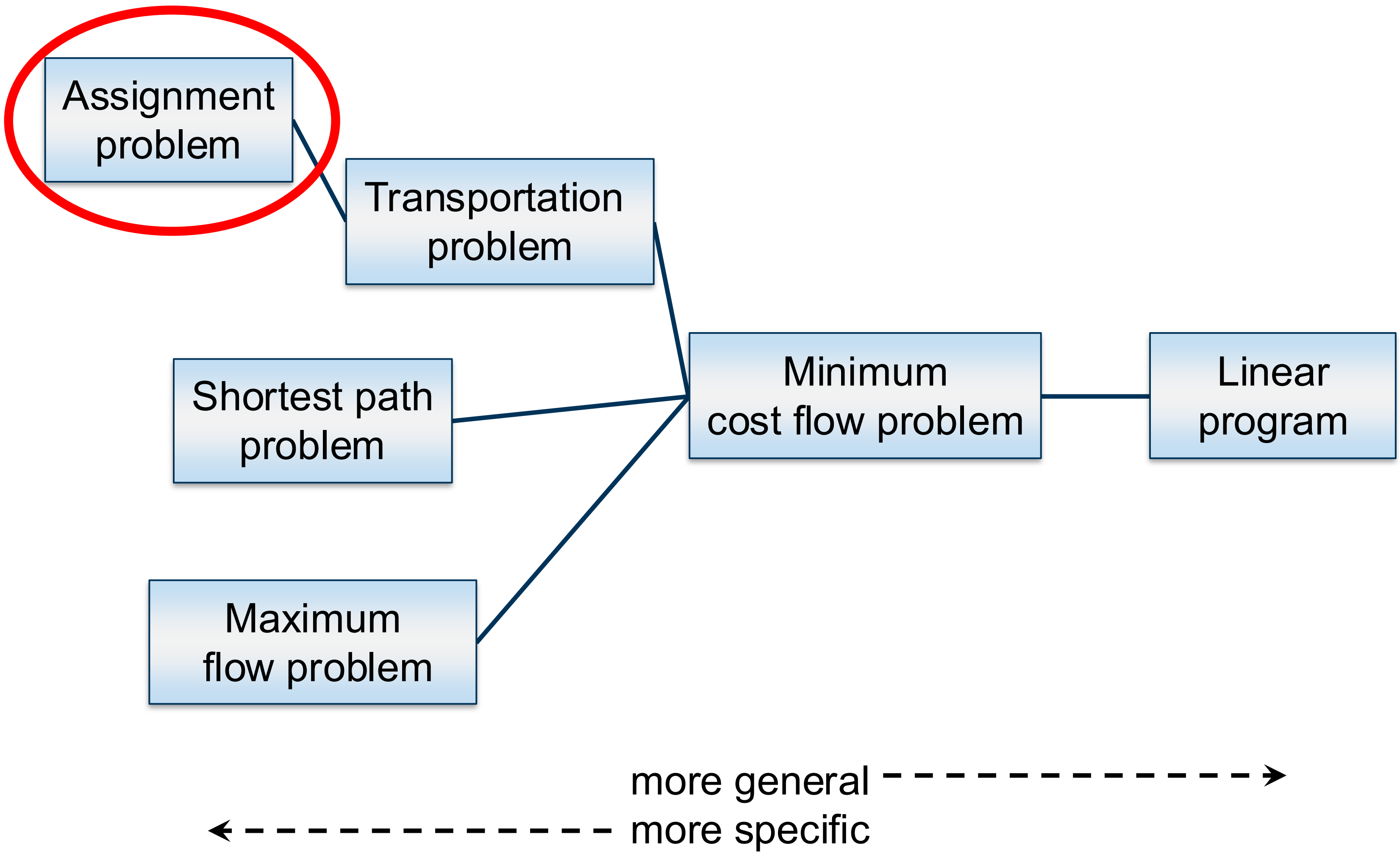
- **Integer Linear Optimization**
  - Modeling integer programming problems (IPs)
  - Solving IPs: branch & bound, cutting planes
  - Advanced solution methods: column generation, approximation algorithms
  - Fast solvable integer problems: total unimodularity, matroids
  - **Network flow problems**
  - Traveling salesperson problem

- **Nonlinear Optimization**
  - Optimization of multidimensional functions
  - Convex optimization

Today you will learn ...

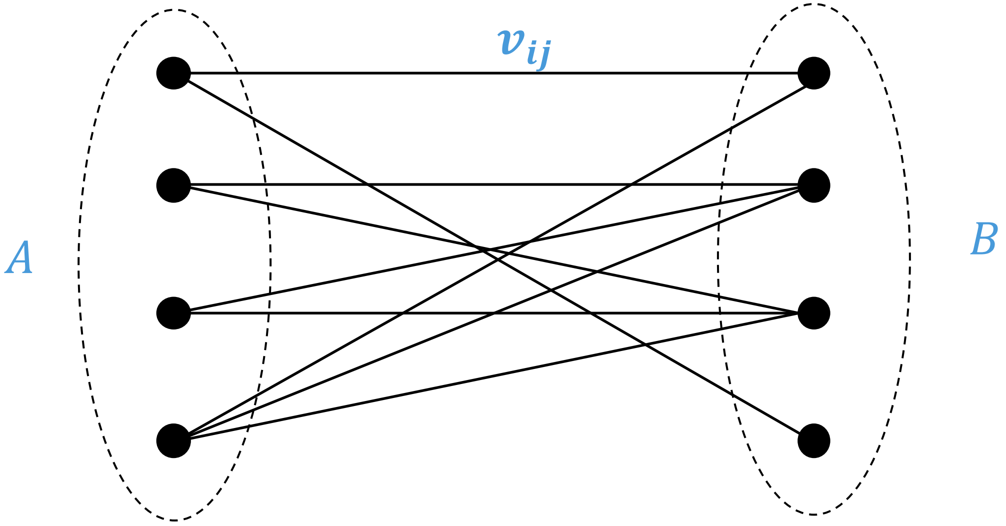
- to model **network flow problems**
- to solve the assignment problem using the **Hungarian algorithm**
- to interpret **shortest path** computation as a network flow problem
- to calculate maximum flows via **Ford-Fulkerson**

Network flow problems with efficient solutions



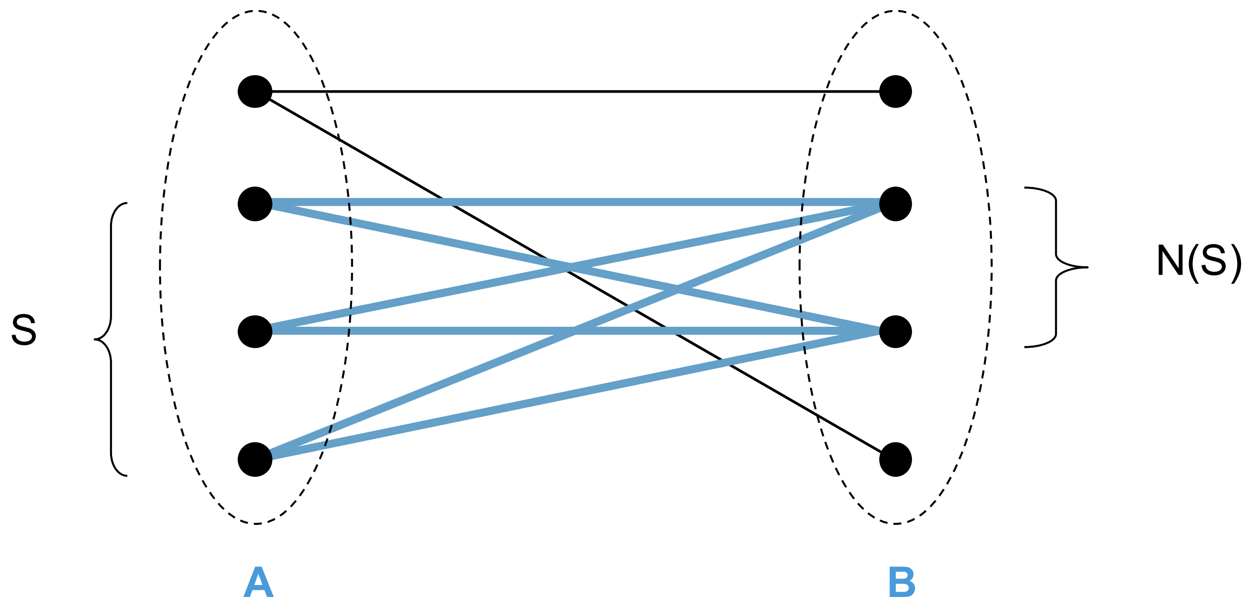
Assignment problem as bipartite matching

- A graph is bipartite if its vertex set can be partitioned into two subsets A and B so that each edge has one vertex in A and the other vertex in B.
- A matching M is a subset of edges, so that every vertex has a degree of at most 1 w.r.t. M.
- The **maximum weight bipartite matching problem**: Find a matching with the maximum weight of edges.



Hall‘s Theorem (1935)

- A bipartite graph $G = (A, B; E)$ has a **perfect matching** that “saturates” A if and only if $|N(S)| \geq |S|$ for every subset $S \subseteq A$, where $N(S) \subseteq B$ are all nodes in B neighbored to S
- A perfect matching matches all vertices of the graph.



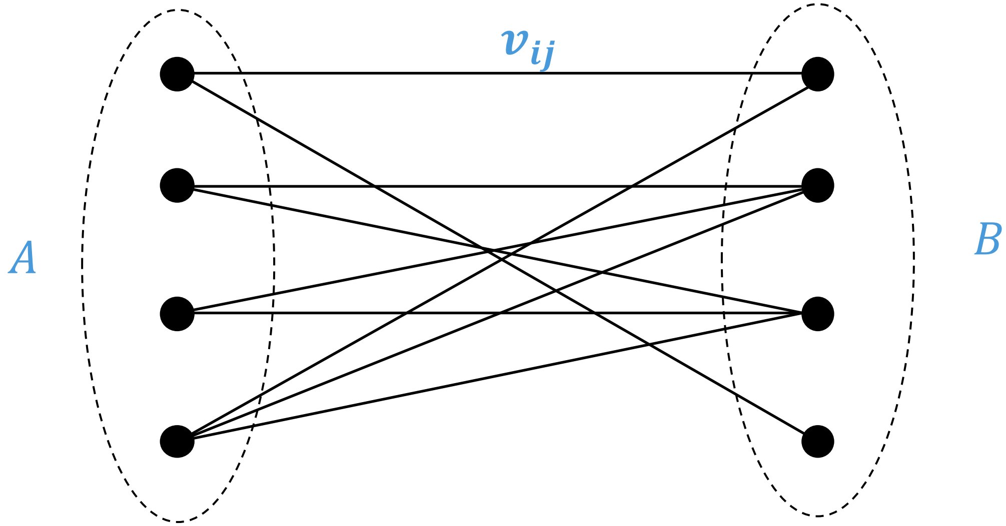
Sufficient conditions for total unimodularity

- A matrix A is totally unimodular, if
  - $a_{ij} \in \{+1, -1, 0\}$ for all i, j
  - Each column has two non-zero coefficients at most
  - There is a partition (M_1, M_2) for the set M of rows, such that
    - If two non-zero entries in a column have the same sign, then the row of one entry is in M_1 the row of the other entry in M_2
    - If two non-zero entries in a column have different signs, then the respective rows are either both in M_1 or in M_2

Assignment problem LP relaxation

$$\max \sum_i \sum_j v_{ij} x_{ij}$$

$$s.t.\ \sum_j x_{ij} \le 1\ \forall\, i \in A$$
$$\sum_i x_{ij} \le 1\ \forall\, j \in B$$
$$x_{ij} \ge 0\ \forall\, i, j$$



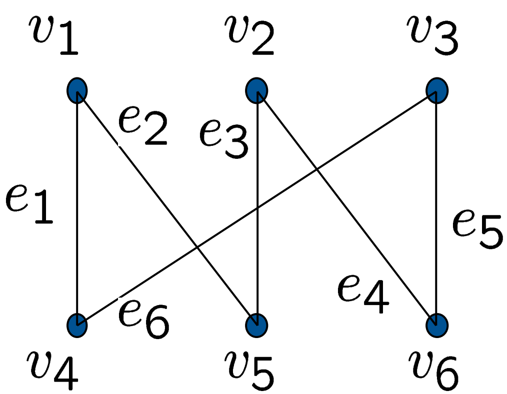
Totally unimodular constraint matrices

Each column of A has exactly 2x the entry “1”.

We can describe A as: $A = \begin{pmatrix} M_1 \\ M_2 \end{pmatrix}$

In a bipartite graph, each column (i.e. a vertex in the graph) of the incidence matrix has a 1 in M_1 and a 1 in $M_{2.}$

The solution to the LP relaxation of the assignment problem is integral.



$$
\begin{array}{cc}
 & \begin{array}{cccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \end{array} \\
\begin{array}{c} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{array} &
\left(\begin{array}{cccccc}
1 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 1 \\
1 & 0 & 0 & 0 & 0 & 1 \\
0 & 1 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 1 & 0
\end{array}\right)
\begin{array}{l}
\left.\begin{array}{c} \\ \\ \\ \end{array}\right\} \boxed{M_1} \\
\left.\begin{array}{c} \\ \\ \\ \end{array}\right\} \boxed{M_2}
\end{array}
\end{array}
$$

An example using the Hungarian algorithm

3 projects should be distributed among 4 contractors.

Each contractor should work on at most one project and each project may be worked on by at most one contractor.

Contractors charge different amounts of money depending on the project.

What is the minimum cost allocation? $m!$ possible allocations

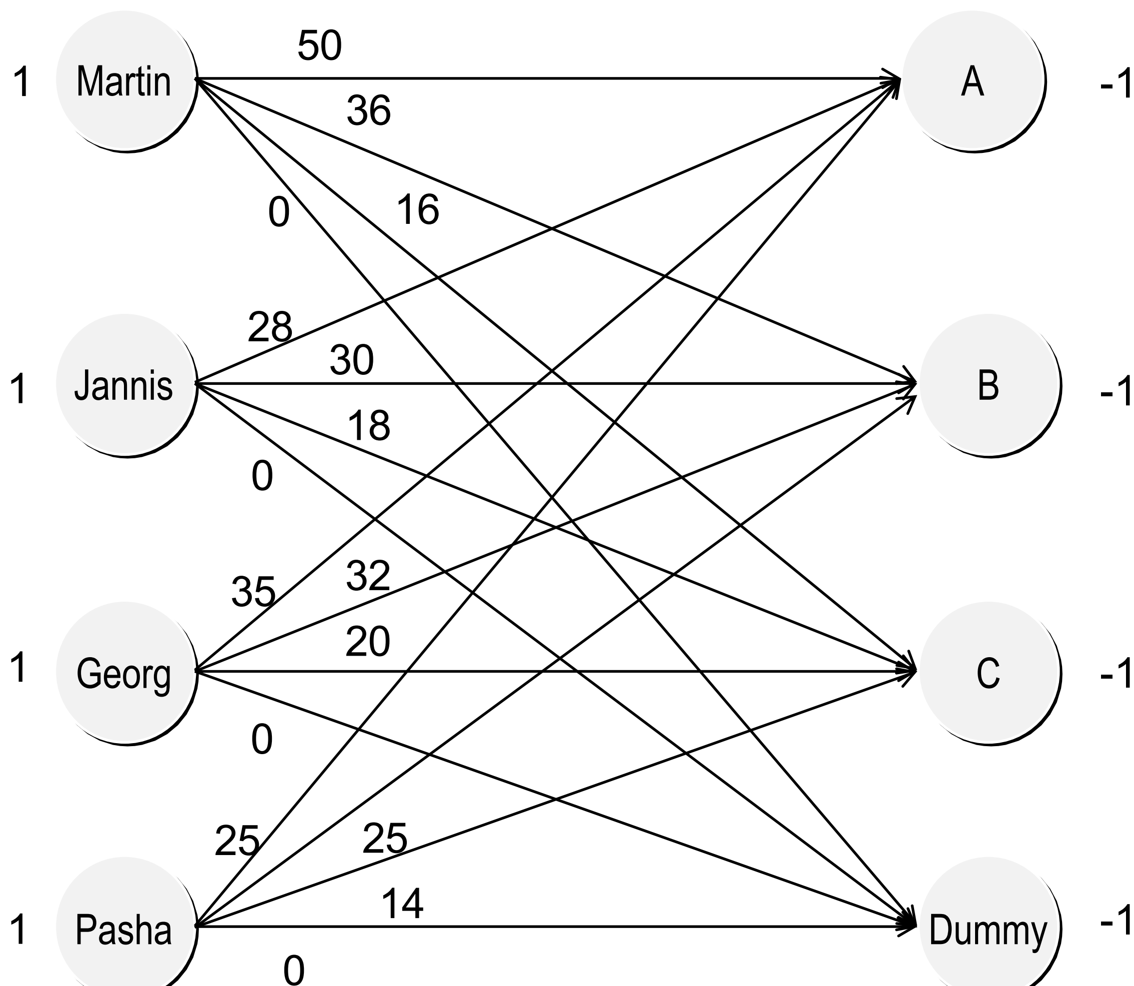
	A	B	C
Martin	50	36	16
Jannis	28	30	18
Georg	35	32	20
Pasha	25	25	14

Dummy project

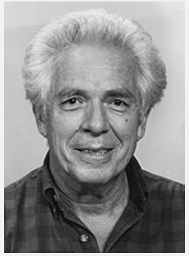
There are 4 contractors and three projects.
We introduce a dummy project that will be assigned to one of the contractors.

	A	B	C	Dummy
Martin	50	36	16	0
Jannis	28	30	18	0
Georg	35	32	20	0
Pasha	25	25	14	0

Network of the assignment problem



The Hungarian method



Published in Naval Research Logistics 1955 by Harold Kuhn, based on work by Dénes Kőnig and Jenő Egerváry.
Later it was found that Carl Gustav Jacobi solved the problem already in the 19th century.
Source: *de.wikipedia.org*

Input: Quadratic $m \times m$ cost matrix

Step 1: Subtract the smallest number in each row from all entries in that row (this guarantees at least one 0 in each row).

Step 2: Subtract the smallest number in each column all entries in that column (this guarantees at least a 0 in each column).

Step 3: Consider the new matrix and compute the smallest number of rows and columns that should be deleted from the matrix such that all zeros are removed.

Step 4: If this number is equal to m, then an optimal solution can be read from the deleted zeros.

Step 5: Consider the smallest true positive entry in the non-deleted rows and columns of the matrix (value k). Subtract k from each non-deleted entry and add k to each entry that is in both a striked row and column.

Go to step 3.

The Hungarian method (example)

Step 1: Subtract the smallest number in each row from all entries in that row.
Because each row contains one zero, the initial matrix is not changed in this step.

	A	**B**	**C**	**Dummy**
Martin	50	36	16	0
Jannis	28	30	18	0
Georg	35	32	20	0
Pasha	25	25	14	0

Step 2: Subtract the smallest number in each column from all entries in that column.

The Hungarian method (example)

Step 2: Subtract the smallest number in each column from all entries in that column.
That results in:

	A	**B**	**C**	**Dummy**
Martin	25	11	2	0
Jannis	3	5	4	0
Georg	10	7	6	0
Pasha	0	0	0	0

The Hungarian method (example)

Step 3: Draw a minimum number of lines to cover all zeros (rows and columns that will be deleted).

	A	B	C	Dummy
Martin	25	11	2	0
Jannis	3	5	4	0
Georg	10	7	6	0
Pasha	0	0	0	0

Step 4: Since the number of lines is smaller than the number of rows $(2 < 4)$, the solution is not optimal.

Step 5: Circle the minimum uncovered number k $(k = 2)$.

- Subtract $k = 2$ from all entries that are not crossed out.
- Add $k = 2$ to all entries where the lines intersect.

The Hungarian method (example)

Step 5:

- Subtract $k = 2$ from all entries that are not crossed out.
- Add $k = 2$ to all entries where the lines intersect.

.

	A	B	C	Dummy
Martin	23	9	0	0
Jannis	1	3	2	0
Georg	8	5	4	0
Pasha	0	0	0	2

The Hungarian method (example)

Step 3: Draw a minimum number of lines to cover all zeros.

	A	B	C	Dummy
Martin	23	9	0	0
Jannis	1	3	2	0
Georg	8	5	4	0
Pasha	0	0	0	2

Step 4: Since the number of lines $3 < 4$ (number of rows), the solution is not yet optimal.

Step 5: Find the smallest uncovered number $k = 1$.

The Hungarian method (example)

Step 5:

- Subtract $k = 1$ from all entries that are not crossed out.
- Add 1 to all entries where the lines intersect.

	A	B	C	Dummy
Martin	23	9	0	1
Jannis	0	2	1	0
Georg	7	4	3	0
Pasha	0	0	0	3

The Hungarian method (example)

Step 3: Draw a minimum number of lines to cover all zeros.

	A	B	C	Dummy
Martin	~~23~~	~~9~~	~~0~~	~~1~~
Jannis	~~0~~	~~2~~	~~1~~	~~0~~
Georg	~~7~~	~~4~~	~~3~~	~~0~~
Pasha	~~0~~	~~0~~	~~0~~	~~3~~

Step 4: Since number of lines 4 = number of rows, the solution is optimal.

The Hungarian method (example)

The optimal assignment is marked by the position of the zeros.
Now there is at least one zero in each line.

	A	B	C	Dummy
Martin	23	9	**0**	1
Jannis	**0**	2	1	0
Georg	7	4	3	**0**
Pasha	0	**0**	0	3

The Hungarian method (example)

The optimal assignment has:

- total costs 69

- There can be several optimal solutions.

	A	B	C	Dummy
Martin	50	36	**16**	0
Jannis	**28**	30	18	0
Georg	35	32	20	**0**
Pasha	25	**25**	14	0

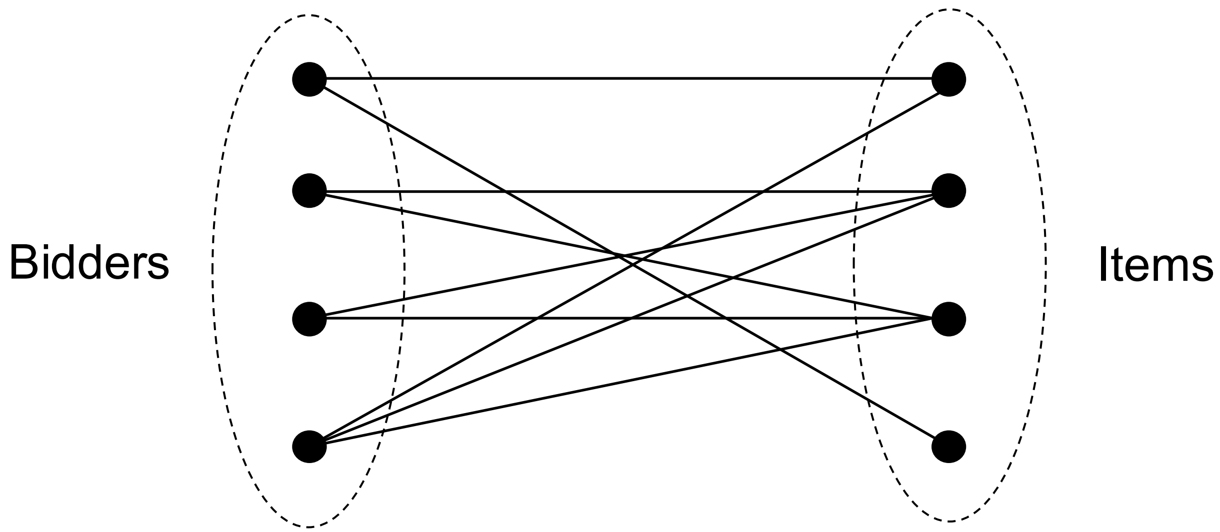
Hungarian Method

- because ideas go back to König and Egerváry (Hungarian mathematicians)
- Running time $O(n^3)$

Contractor	Project	Cost
Martin	C	16
Jannis	A	28
Georg	Not assigned	
Pasha	B	25

Assignment markets – the assignment problem as an auction

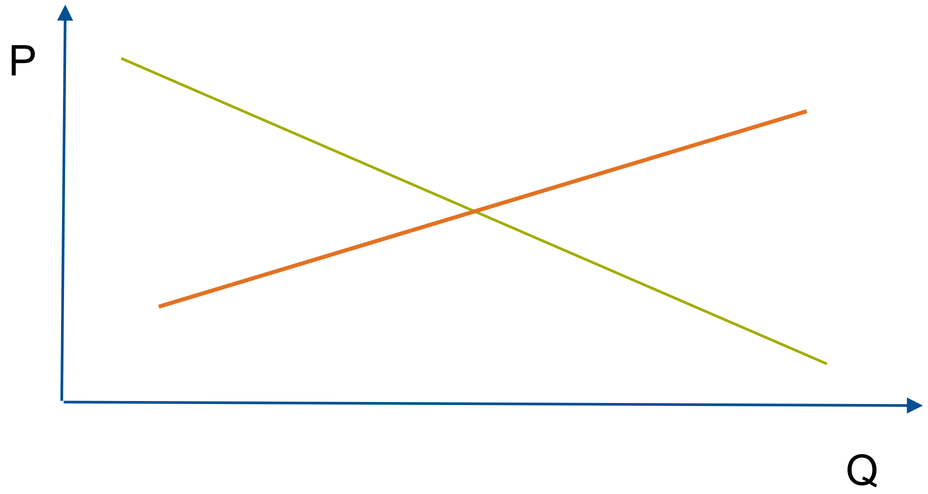
- Each bidder wants to win **at most 1 item** and 1 item can be **assigned to at most 1 bidder**.



Competitive equilibrium

We‘re looking for a **competitive equilibrum** price vector:

- Buyers maximize their utility at these prices
- Sellers maximize their profits at these prices.
- The market is budget balanced (no surplus or subsidy required).



How to find (minimal) competitive equilibrium prices?

We formulate the assignment game as a linear program with the following variables:

v_{ij} – value of a bidder i for item j

π_i – profit of bidder i; p_j – price for item j

Primal Program

$$Max\ \Sigma_i \Sigma_j v_{ij} x_{ij}$$

$$s.t.\ \Sigma_j x_{ij} = 1\ \forall\ i\ Bidders$$

$$\Sigma_i x_{ij} = 1\ \forall j\ Items$$

$$x_{ij} \geq 0\ \forall\ i, j$$

Dual Program

$$Min\ \Sigma_i \pi_i + \Sigma_j p_j$$

$$s.t.\ \pi_i + p_j \geq v_{ij} \quad \forall\ i, j$$

$$\pi_i, p_j\ unrestricted$$

Primal complementary slackness:

$$\pi_i + p_j > v_{ij} \Rightarrow x_{ij} = 0$$

Dual by Leonard (1983)

$$Min\ \Sigma_j p_j$$

$$s.t.\ \pi_i + p_j \geq v_{ij} \quad \forall\ i, j$$

$$\Sigma \pi_i + \Sigma p_j = W^*$$

$$\pi_i, p_j \geq 0$$

The DGS auction by Demange et al. (1986)

A set of items is minimally overdemanded, if the number of bidders demanding only items in this set is greater than the number of items in this set.
Minimally, means that no proper subset of the items is overdemanded.

Items

Round 1 *Bidders* $\begin{pmatrix} 5 & \mathbf{7} & 2' \\ 8' & \mathbf{9} & 3 \\ 2 & \mathbf{6}' & 0 \end{pmatrix}$

Prices $(0 \quad 1 \quad 0)$

Round 2 $\begin{pmatrix} 5 & \mathbf{6} & 2 \\ \mathbf{8} & \mathbf{8} & 3 \\ 2 & \mathbf{5} & 0 \end{pmatrix}$

Prices $(0 \quad 2 \quad 0)$

Round 3 $\begin{pmatrix} \mathbf{5} & \mathbf{5} & 2 \\ \mathbf{8} & 7 & 3 \\ 2 & \mathbf{4} & 0 \end{pmatrix}$

Prices $(1 \quad 3 \quad 0)$

Round 4 $\begin{pmatrix} \mathbf{4} & \mathbf{4} & 2 \\ \mathbf{7} & 6 & 3 \\ 1 & \mathbf{3} & 0 \end{pmatrix}$

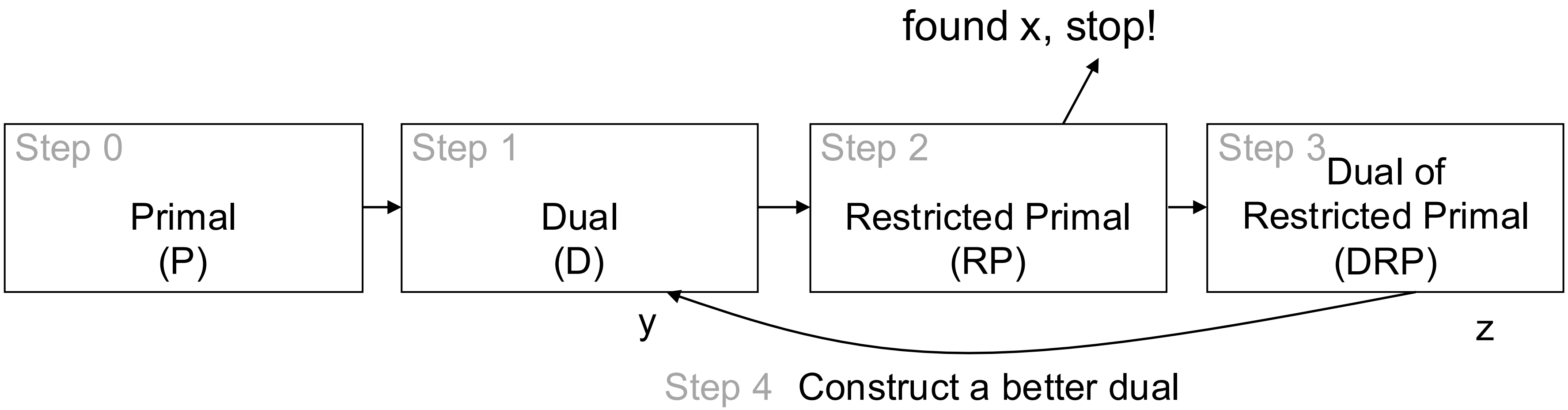
Prices $(2 \quad 4 \quad 0)$

Round 5 $\begin{pmatrix} \mathbf{3} & \mathbf{3} & 2 \\ \mathbf{6} & 5 & 3 \\ 0 & \mathbf{2} & 0 \end{pmatrix}$

Prices $(3 \quad 5 \quad 0)$

Round 6 $\begin{pmatrix} \mathbf{2} & \mathbf{2} & \mathbf{2} \\ \mathbf{5} & 4 & 3 \\ -1 & \mathbf{1} & 0 \end{pmatrix}$

Primal-dual algorithms for solving LPs



The Hungarian algorithm is a precursor of the primal-dual method for solving LPs, which is often used to construct approximation algorithms.

$$Max\ \Sigma\, \Sigma v_{ij} x_{ij}$$

$$s.t.\ \Sigma x_{ij} = 1\ \forall\, i$$
$$\Sigma x_{ij} = 1\ \forall j$$
$$x_{ij} \geq 0\ \forall\, i, j$$

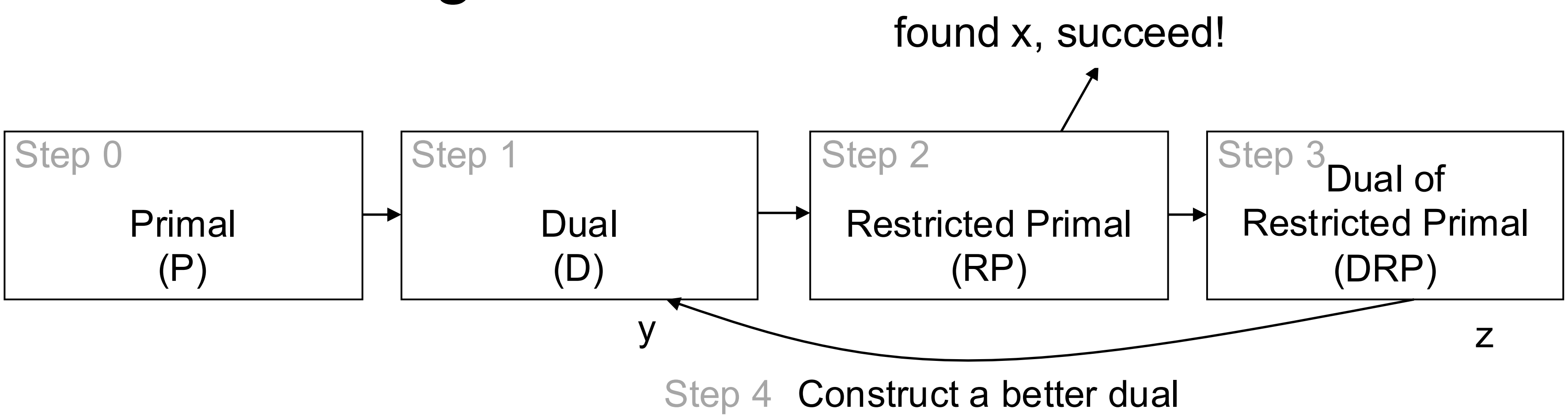
$$Min\ \Sigma \pi_i + \Sigma p_j$$

$$s.t.\ \pi_i + p_j \geq v_{ij} \quad \forall\, i, j$$

π_i, p_j unlimited

Complementary slack: $\pi_i + p_j > v_{ij} \Rightarrow x_{ij} = 0$

Primal-dual algorithms and auctions



- **Step 0:** At the beginning, there is a (maximum) assignment problem, i.e., a relaxed primal LP.
- **Step 1:** The dual problem can be derived.
- **Step 2:** By using complementary slackness conditions, a second primal problem can be formulated ➔ restricted (by a dual feasible solution) primal problem (RP)
- If the value of the restricted primal problem is zero, you will get an optimal solution (for the original primal problem) .
- **Step 3:** Otherwise, the dual of the restricted problem (DRP) is formulated.
- **Step 4:** Update to better duals
- Repeat the process between RP and DRP until the RP has a solution with value of zero!

Start Example as an LP – 1st Iteration, Step 0 (P)

$$\begin{array}{cc} & \text{Items} \\ \text{Bidders} & \begin{pmatrix} 5 & \mathbf{7} & 2 \\ 8 & \mathbf{9} & 3 \\ 2 & \mathbf{6} & 0 \end{pmatrix} \end{array}$$

$$\begin{aligned} & Max \; \Sigma_i \; \Sigma_j v_{ij} x_{ij} \\ s.t. \; & \Sigma_j x_{ij} = 1 \;\forall\, i \text{ Bidders} \\ & \Sigma_i x_{ij} = 1 \;\forall j \text{ Items} \\ & x_{ij} \geq 0 \;\forall\, i, j \end{aligned}$$

Primal Program

In our example the relaxed **primal LP (P)** is:

$$Max\ Z = 5x_{1A} + 7x_{1B} + 2x_{1C} + 8x_{2A} + 9x_{2B} + 3x_{2C} + 2x_{3A} + 6x_{3B} + 0x_{3C}$$

$$s.t.$$

$$x_{1A} + x_{1B} + x_{1C} = 1 \qquad (\pi_1)$$

$$x_{2A} + x_{2B} + x_{2C} = 1 \qquad (\pi_2)$$

$$x_{3A} + x_{3B} + x_{3C} = 1 \qquad (\pi_3)$$

$$x_{1A} + x_{2A} + x_{3A} = 1 \qquad (p_A)$$

$$x_{1B} + x_{2B} + x_{3B} = 1 \qquad (p_B)$$

$$x_{1C} + x_{2C} + x_{3C} = 1 \qquad (p_C)$$

$$x_{ij} \geq 0 \quad \forall\, i = 1,2,3 \;\; \& \;\; j = A, B, C$$

Example – 1st Iteration, Step 1 (D)

1. Formulate the dual LP (D):

$$
\begin{aligned}
Min\ & \pi_1 + \pi_2 + \pi_3 + p_A + p_B + p_C \\
s.t.\ & \pi_1 + p_A \geq 5 \qquad (x_{1A}) \\
& \pi_1 + p_B \geq 7 \qquad (x_{1B}) \\
& \pi_1 + p_C \geq 2 \qquad (x_{1C}) \\
& \pi_2 + p_A \geq 8 \qquad (x_{2A}) \\
& \pi_2 + p_B \geq 9 \qquad (x_{2B}) \\
& \pi_2 + p_C \geq 3 \qquad (x_{2C}) \\
& \pi_3 + p_A \geq 2 \qquad (x_{3A}) \\
& \pi_3 + p_B \geq 6 \qquad (x_{3B}) \\
& \pi_3 + p_C \geq 0 \qquad (x_{3C}) \\
& \pi_i, p_j \text{ unrestricted}
\end{aligned}
$$

$$
\begin{aligned}
& Min\ \Sigma\pi_i + \Sigma p_j \\
& s.t.\ \pi_i + p_j \geq v_{ij} \quad \forall\, i,j \\
& \pi_i, p_j\ unrestricted
\end{aligned}
$$

Primal complementary slackness:

$$\pi_i + p_j > v_{ij} \Rightarrow x_{ij} = 0$$

The following x_{ij} are in J and ≥ 0:

$$\{x_{1B}, x_{2B}, x_{3B}\}$$

⇨ All other primal variables are zero!

Solution with feasible starting prices of 0:

$p = (0,0,0) \quad \& \quad \pi = (7,9,6)$

Example - 1st Iteration, Step 2 (RP)

1. Formulate the (RP) and solve it:

$$
\begin{aligned}
&Max\ \ Z = -s_1 - s_2 - s_3 - t_B \\
&\text{s.t.} \quad x_{1B} + s_1 = 1 \quad (\pi_1) \\
&\quad x_{2B} + s_2 = 1 \quad (\pi_2) \\
&\quad x_{3B} + s_3 = 1 \quad (\pi_3) \\
&\quad x_{1B} + x_{2B} + x_{3B} + t_B = 1 \quad (p_B)
\end{aligned}
$$

$$x_{ij} \geq 0\ \forall (i,j) \in J$$

$$x_{ij} = 0\ \forall (i,j) \in E \setminus J$$

$$s_1, s_2, s_3, t_B \geq 0$$

$Max. -\Sigma s_i - \Sigma t_j$

$s.t.\ \Sigma x_{ij} + s_i = 1\ \forall i$ active

$\Sigma x_{ij} + t_j = 1\ \forall j$ demanded

$s_i, t_j, x_{ij} \geq 0$

$x_{ij} = 0\ \forall (i,j) \in E \setminus J$

$J = \{(i,j) \in E : \pi_i + p_j = v_{ij}\}$

Here $J = \{(1,B); (2,B); (3,B)\}$

- This is the same problem as trying to find an assignment in the resulting subgraph.
- If $Z = 0$, then terminate.
- s are unsatisfied bidders, while t are unallocated items in the demand sets of the bidders.
- This RP has $Z = -2$, which is the number of free nodes in the matching, i.e., the number of unsatisfied bidders.

1^{st} iteration, Step 3 (DRP) + Step 4 (Dual Variables Update)

1. Formulate the dual of the restricted primal (DRP):

$$
\begin{aligned}
& Min\ Z = \pi_1' + \pi_2' + \pi_3' + p_B' \\
\text{s.t.}\quad & \pi_1' + p_B' \geq 0 \quad (x_{1B}) \\
& \pi_2' + p_B' \geq 0 \quad (x_{2B}) \\
& \pi_3' + p_B' \geq 0 \quad (x_{3B}) \\
& \pi_1' \geq -1 \quad (s_1) \\
& \pi_2' \geq -1 \quad (s_2) \\
& \pi_3' \geq -1 \quad (s_3) \\
& p_B' \geq -1 \quad (t_B)
\end{aligned}
$$

$$
\begin{aligned}
& Min\ \Sigma\, \pi_i' + \Sigma p_j' \\
& s.t.\ \pi_i' + p_j' \geq 0\ \ \forall\ (i,j) \in J \\
& \pi_i' \geq -1\ \forall\ i \text{ active} \\
& p_j' \geq -1\ \forall\ j \text{ minimally overdemanded} \\
& p_j' \geq 0\ \forall\ j \text{ else}
\end{aligned}
$$

2. Find optimal solution to the DRP:
   1. It can be shown that every basic solution has all its components equal to +1 or -1;
   2. Here, $p' = (0{,}1{,}0)$ is the direction of price change for item j and $\pi' = (-1,-1,-1)$ the change in bidder i's payoff, $Z = -3 + 1 = -2$.

3. Update dual variables:

$p^{new} = (0{,}0{,}0) + (0{,}1{,}0) = (0{,}1{,}0)$

$\pi^{new} = (7{,}9{,}6) + (-1,-1,-1) = (6{,}8{,}5)$

$$
\begin{pmatrix} 5 & \mathbf{7} & 2 \\ 8 & \mathbf{9} & 3 \\ 2 & \mathbf{6} & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 5 & \mathbf{6} & 2 \\ \mathbf{8} & \mathbf{8} & 3 \\ 2 & \mathbf{5} & 0 \end{pmatrix}
$$

Example – 2nd Iteration, Step 1 (D)

1. Z not 0, so formulate the dual LP (D):

$$
\begin{aligned}
&Min\ Z = \pi_1 + \pi_2 + \pi_3 + p_A + p_B + p_C \\
&s.t. \quad \pi_1 + p_A \geq 5 \quad (x_{1A}) \\
&\quad \pi_1 + p_B \geq 7 \quad (x_{1B}) \\
&\quad \pi_1 + p_C \geq 2 \quad (x_{1C}) \\
&\quad \pi_2 + p_A \geq 8 \quad (x_{2A}) \\
&\quad \pi_2 + p_B \geq 9 \quad (x_{2B}) \\
&\quad \pi_2 + p_C \geq 3 \quad (x_{2C}) \\
&\quad \pi_3 + p_A \geq 2 \quad (x_{3A}) \\
&\quad \pi_3 + p_B \geq 6 \quad (x_{3B}) \\
&\quad \pi_3 + p_C \geq 0 \quad (x_{3C}) \\
&\quad \pi_i, p_j \text{ unrestricted}
\end{aligned}
$$

- Updated solution: $p = (0,1,0)$ & $\pi = (6,8,5)$

$$
\begin{gathered}
Min\ \Sigma\pi_i + \Sigma p_j \\
s.t.\ \pi_i + p_j \geq v_{ij} \quad \forall\, i,j \\
\pi_i, p_j\ unrestricted
\end{gathered}
$$

Primal complementary slackness:

$$\pi_i + p_j > v_{ij} \Rightarrow x_{ij} = 0$$

The following x_{ij} are in J and ≥ 0:

$$\{x_{1B}, x_{2A}, x_{2B}, x_{3B}\}$$

All other primal variables are zero!

Example – 2nd Iteration, Step 2 (RP)

1. Formulate the (RP) and solve it:

$$
\begin{aligned}
&Max\ Z = -s_1 - s_2 - s_3 - t_A - t_B \\
&s.t. \qquad x_{1B} + s_1 = 1 \quad (\pi_1) \\
&x_{2A} + x_{2B} + s_2 = 1 \quad (\pi_2) \\
&x_{3B} + s_3 = 1 \quad (\pi_3) \\
&x_{2A} + t_A = 1 \quad (p_A) \\
&x_{1B} + x_{2B} + x_{3B} + t_B = 1 \quad (p_B)
\end{aligned}
$$

$$x_{ij} \geq 0\ \forall (i,j) \in J$$

$$x_{ij} = 0\ \forall (i,j) \in E \backslash J$$

$$Max\ Z = -\Sigma s_i - \Sigma t_j$$

$$s.t.\ \Sigma x_{ij} + s_i = 1\ \forall i \textit{ active}$$

$$\Sigma x_{ij} + t_j = 1\ \forall j \textit{ demanded}$$

$$s_i, t_j, x_{ij} \geq 0$$

$$x_{ij} = 0\ \forall (i,j) \in E \backslash J$$

$$J = \{(i,j) \in E: \pi_i + p_j = v_{ij}\}$$

Here $J = \{(1,B); (2,A); (2,B); (3,B)\}$

- This is the same problem as trying to find an assignment in the resulting subgraph.
- If $Z = 0$, then terminate.
- s are unsatisfied bidders, while t are unallocated items in the demand sets of the bidders.
- This RP has $Z = -1$, i.e., now there is only 1 unsatisfied bidder.

2nd Iteration, Step 3 (DRP) + Step 4 (Dual Variables Update)

- Formulate the dual of the restricted primal (DRP):

$$
\begin{array}{llr}
\mathit{Min}\ Z = \pi_1' + \pi_2' + \pi_3' + p_A' + p_B' & & \\
s.t. \quad \pi_1' + p_B' \geq 0 & & (x_{1B}) \\
\pi_2' + p_A' \geq 0 & & (x_{2A}) \\
\pi_2' + p_B' \geq 0 & & (x_{2B}) \\
\pi_3' + p_B' \geq 0 & & (x_{3B}) \\
\pi_1' \geq -1 & & (s_1) \\
\pi_2' \geq -1 & & (s_2) \\
\pi_3' \geq -1 & & (s_3) \\
p_A' \geq -1 & & (t_A) \\
p_B' \geq -1 & & (t_B)
\end{array}
$$

$$\mathit{Min}\ \Sigma\, \pi_i' + \Sigma p_j'$$
$$s.t.\ \pi_i' + p_j' \geq 0\ \ \forall\ (i,j) \in J$$
$\pi_i' \geq -1\ \forall\ i$ *active*

$p_j' \geq -1\ \forall\ j$ *minimally overdemanded*

$p_j' \geq 0\ \forall\ j$ *else*

- Find optimal solution to the DRP: $p' = (0,1,0)$ is the direction of price change for item j and $\pi' = (-1,0,-1)$ the change in bidder i's payoff, $Z = -2 + 1 = -1$.
- Update dual variables:

$$p^{new} = (0,1,0) + (0,1,0) = (0,2,0)$$
$$\pi^{new} = (6,8,5) + (-1,0,-1) = (5,8,4)$$

$$\begin{pmatrix} 5 & \mathbf{6} & 2 \\ \mathbf{8} & \mathbf{8} & 3 \\ 2 & \mathbf{5} & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} \mathbf{5} & \mathbf{5} & 2 \\ \mathbf{8} & 7 & 3 \\ 2 & \mathbf{4} & 0 \end{pmatrix}$$

Example – 3rd Iteration, Step 1 (D)

1. Z not 0, so formulate the dual LP (D):

$$Min\ Z = \pi_1 + \pi_2 + \pi_3 + p_A + p_B + p_C$$

s.t.

$$\begin{array}{rl}
\pi_1 + p_A \geq 5 & (x_{1A}) \\
\pi_1 + p_B \geq 7 & (x_{1B}) \\
\pi_1 + p_C \geq 2 & (x_{1C}) \\
\pi_2 + p_A \geq 8 & (x_{2A}) \\
\pi_2 + p_B \geq 9 & (x_{2B}) \\
\pi_2 + p_C \geq 3 & (x_{2C}) \\
\pi_3 + p_A \geq 2 & (x_{3A}) \\
\pi_3 + p_B \geq 6 & (x_{3B}) \\
\pi_3 + p_C \geq 0 & (x_{3C})
\end{array}$$

π_i, p_j unrestricted

$$Min\ \Sigma\pi_i + \Sigma p_j$$
$$s.t.\ \pi_i + p_j \geq v_{ij} \quad \forall\, i,j$$
$$\pi_i, p_j\ unrestricted$$

Primal complementary slackness:

$$\pi_i + p_j > v_{ij} \Rightarrow x_{ij} = 0$$

The following x_{ij} are in J and ≥ 0:

$\{x_{1A}, x_{1B}, x_{2A}, x_{3B}\}$

Updated solution: $p = (0,2,0)$ & $\pi = (5,8,4)$

Example – 3rd Iteration, Step 2 (RP)

1. Formulate the (RP) and solve it:

$$
\begin{aligned}
&Max\ Z = -s_1 - s_2 - s_3 - t_A - t_B \\
&s.t.\ \ x_{1A} + x_{1B} + s_1 = 1 \qquad (\pi_1) \\
&x_{2A} + s_2 = 1 \qquad (\pi_2) \\
&x_{3B} + s_3 = 1 \qquad (\pi_3) \\
&x_{1A} + x_{2A} + t_A = 1 \qquad (p_A) \\
&x_{1B} + x_{3B} + t_B = 1 \qquad (p_B)
\end{aligned}
$$

$$
\begin{aligned}
x_{ij} &\geq 0\ \forall (i,j) \in J \\
x_{ij} &= 0\ \forall (i,j) \in E/J
\end{aligned}
$$

$Max\ Z = -\Sigma s_i - \Sigma t_j$

$s.t.\ \Sigma x_{ij} + s_i = 1\ \forall i$ *active*

$\Sigma x_{ij} + t_j = 1\ \forall j$ *demanded*

$s_i, t_j, x_{ij} \geq 0$

$x_{ij} = 0\ \forall (i,j) \in E \setminus J$

$J = \{(i,j) \in E: \pi_i + p_j = v_{ij}\}$

Here: $J = \{(1,A); (1,B); (2,A); (3,B)\}$

- This is the same problem as trying to find an assignment in the resulting subgraph.
- If $z = 0$, then terminate.
- s are unsatisfied bidders, while t are unallocated items in the demand sets of the bidders.
- This RP has $Z = -1$, i.e., now there is still 1 unsatisfied bidder.

3rd Iteration, Step 3 (DRP) + Step 4 (dual variables update)

1. Formulate the dual of the restricted primal (DRP):

$$\begin{aligned}
&Min\ Z = \pi_1' + \pi_2' + \pi_3' + p_A' + p_B' \\
&s.t. \quad \pi_1' + p_A' \geq 0 \quad (x_{1B}) \\
&\qquad \pi_1' + p_B' \geq 0 \quad (x_{2A}) \\
&\qquad \pi_2' + p_A' \geq 0 \quad (x_{2B}) \\
&\qquad \pi_3' + p_B' \geq 0 \quad (x_{3B}) \\
&\qquad \pi_1' \geq -1 \quad (s_1) \\
&\qquad \pi_2' \geq -1 \quad (s_2) \\
&\qquad \pi_3' \geq -1 \quad (s_3) \\
&\qquad p_A' \geq -1 \quad (t_A) \\
&\qquad p_B' \geq -1 \quad (t_B)
\end{aligned}$$

$$Min\ \Sigma\, \pi_i' + \Sigma p_j'$$
$$s.t.\ \pi_i' + p_j' \geq 0\ \ \forall\, (i,j) \in J$$
$$\pi_i' \geq -1\ \forall\ i \text{ active}$$
$$p_j' \geq -1\ \forall\ j \text{ minimally overdemanded}$$
$$p_j' \geq 0\ \forall\ j \text{ else}$$

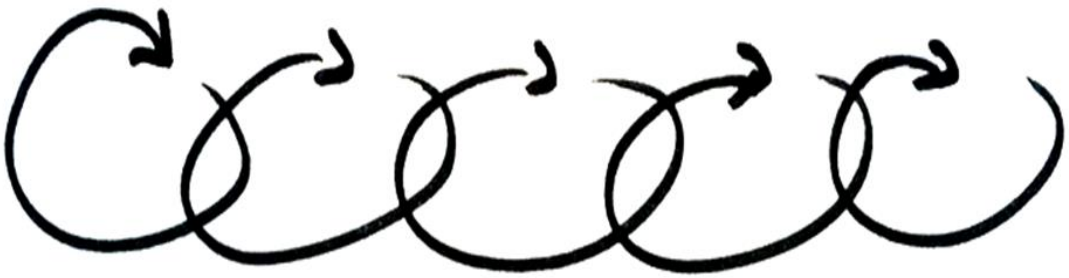
2. Find optimal solution to the DRP: $p' = (1,1,0)$ is the direction of price change for item j and $\pi' = (-1,-1,-1)$ the change in bidder i's payoff, $Z = -3 + 2 = -1$.

3. Update dual variables:

$$p^{new} = (0,2,0) + (1,1,0) = (1,3,0)$$
$$\pi^{new} = (5,8,4) + (-1,-1,-1) = (4,7,3)$$

$$\begin{pmatrix} \mathbf{5} & \mathbf{5} & 2 \\ \mathbf{8} & 7 & 3 \\ 2 & \mathbf{4} & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} \mathbf{4} & \mathbf{4} & 2 \\ \mathbf{7} & 6 & 3 \\ 1 & \mathbf{3} & 0 \end{pmatrix}$$

Iteration 4 - 6



Example – 6th Iteration, Step 1 (D)

1. Z not 0, so formulate the dual LP (D):

$$Min\ Z = \pi_1 + \pi_2 + \pi_3 + p_A + p_B + p_C$$

s.t.
$$\pi_1 + p_A \geq 5 \quad (x_{1A})$$
$$\pi_1 + p_B \geq 7 \quad (x_{1B})$$
$$\pi_1 + p_C \geq 2 \quad (x_{1C})$$
$$\pi_2 + p_A \geq 8 \quad (x_{2A})$$
$$\pi_2 + p_B \geq 9 \quad (x_{2B})$$
$$\pi_2 + p_C \geq 3 \quad (x_{2C})$$
$$\pi_3 + p_A \geq 2 \quad (x_{3A})$$
$$\pi_3 + p_B \geq 6 \quad (x_{3B})$$
$$\pi_3 + p_C \geq 0 \quad (x_{3C})$$

π_i, p_j unrestricted

$$Min\ \Sigma\pi_i + \Sigma p_j$$
$$s.t.\ \pi_i + p_j \geq v_{ij} \quad \forall\, i,j$$
$$\pi_i, p_j\ unrestricted$$

Primal complementary slackness:

$$\pi_i + p_j > v_{ij} \Rightarrow x_{ij} = 0$$

The following x_{ij} are in J and $\neq 0$:
$\{x_{1A}, x_{1B}, x_{1C}, x_{2A}, x_{3B}\}$

Updated solution: $p = (3{,}5{,}0)$ & $\pi = (2{,}5{,}1)$

Example – 6th Iteration, Step 2 (RP)

1. Formulate the (RP) and solve it:

$$Max\ Z = -s_1 - s_2 - s_3 - t_A - t_B - t_C$$

$$\begin{aligned}
s.t.\quad x_{1A} + x_{1B} + x_{1C} + s_1 &= 1 \quad (\pi_1)\\
x_{2A} + s_2 &= 1 \quad (\pi_2)\\
x_{3B} + s_3 &= 1 \quad (\pi_3)\\
x_{1A} + x_{2A} + t_A &= 1 \quad (p_A)\\
x_{1B} + x_{3B} + t_B &= 1 \quad (p_B)\\
x_{1C} + t_C &= 1 \quad (p_c)
\end{aligned}$$

$$x_{ij} \geq 0\ \forall (i,j) \in J$$
$$x_{ij} = 0\ \forall (i,j) \in E \backslash J$$

$$Max\ \mathrm{z} = -\Sigma s_i - \Sigma \mathrm{t}_j$$
$$s.t.\ \Sigma x_{ij} + s_i = 1\ \forall i\ active$$
$$\Sigma x_{ij} + t_j = 1\ \forall j\ demanded$$
$$s_i, t_j, x_{ij} \geq 0$$
$$x_{ij} = 0\ \forall (i,j) \in E \backslash J$$

$$J = \{(i,j) \in E: \pi_i + p_j = v_{ij}\}$$

Here

$$J = (1,A);(1,B);(\mathbf{1},\boldsymbol{C});(\mathbf{2},\boldsymbol{A});(\mathbf{3},\boldsymbol{B})\}$$

- This is the same problem as trying to find an assignment in the resulting subgraph.
- If $z = 0$, then terminate.
- s are unsatisfied bidders, while t are unallocated items in the demand sets of the bidders.

- This RP has $Z = 0$, i.e., now there is no unsatisfied bidder

Termination!

CE of the Assignment Game and LP duality

Theorem: The set of CE of an assignment game is precisely the set of solutions of the LP dual of the corresponding assignment problem (Shapley & Shubik, 1971)

- The theorem follows directly from linear programming duality
- We know that CE exists in this game, and we know effective computational procedures (given that bidders submit sealed bids).

In our example, we had a competitive equilibrium price vector $p = (3{,}5{,}0)$

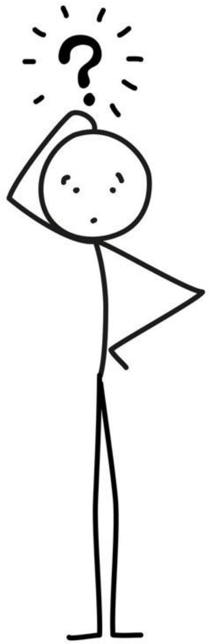
Matrix with bidder valuations:

	A	B	C	π_i
Bidder 1	5	7	$\mathbf{2}^*$	$\mathbf{2}$
Bidder 2	$\mathbf{8}^*$	9	3	$\mathbf{5}$
Bidder 3	2	$\mathbf{6}^*$	0	$\mathbf{1}$
p_j	$\mathbf{3}$	$\mathbf{5}$	$\mathbf{0}$	

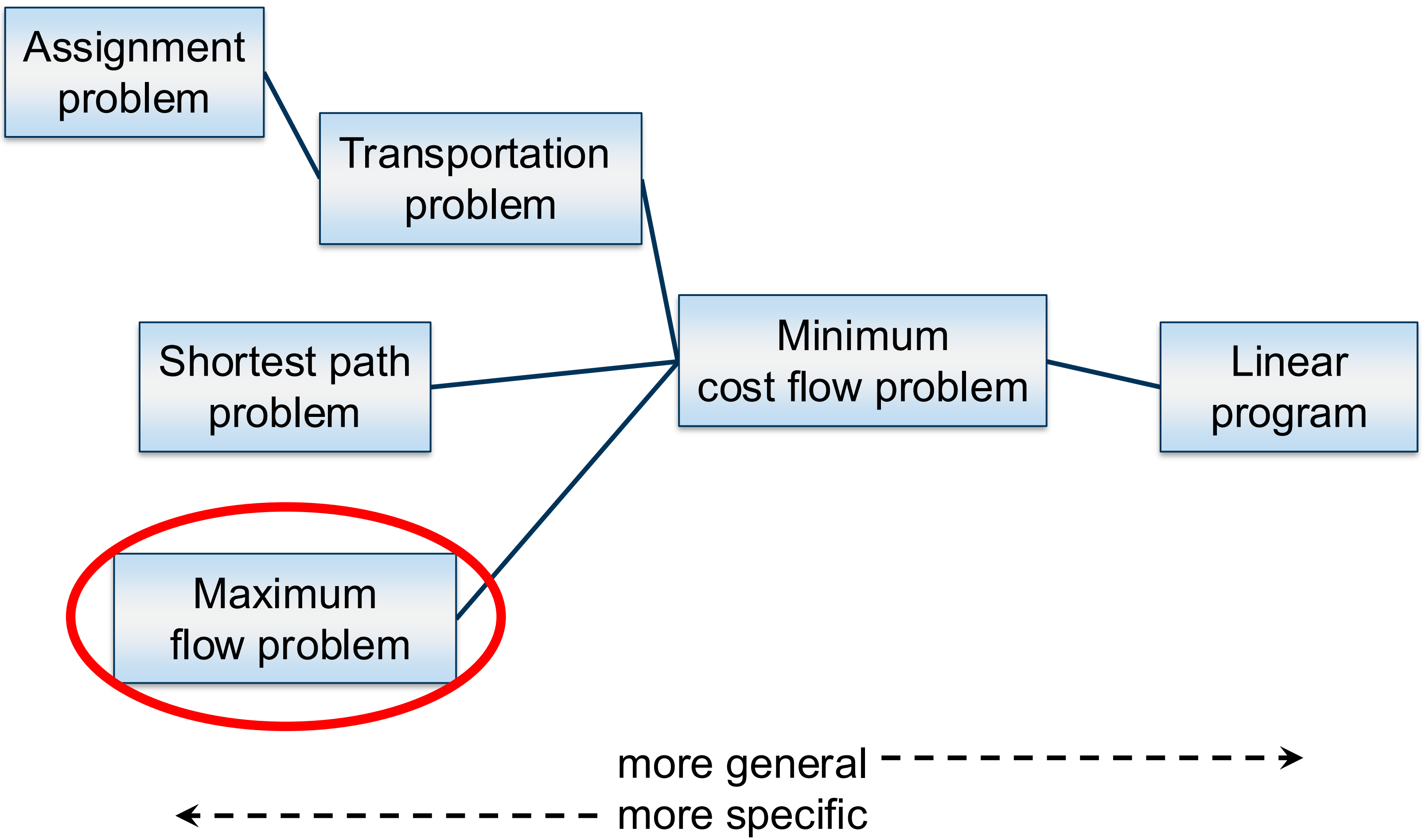
Bidder payoffs:

	Item A	Item B	Item C
B1	$5 - 3 = 2$	$7 - 5 = 2$	$\mathbf{2 - 0 = 2}^*$
B2	$\mathbf{8 - 3 = 5}^*$	$9 - 5 = 4$	$3 - 0 = 3$
B3	$2 - 3 = -1$	$\mathbf{6 - 5 = 1}^*$	$0 - 0 = 0$

What property of network flow problems causes the solutions to always be integer?



Network flow problems with efficient solutions



Maximum flow problem

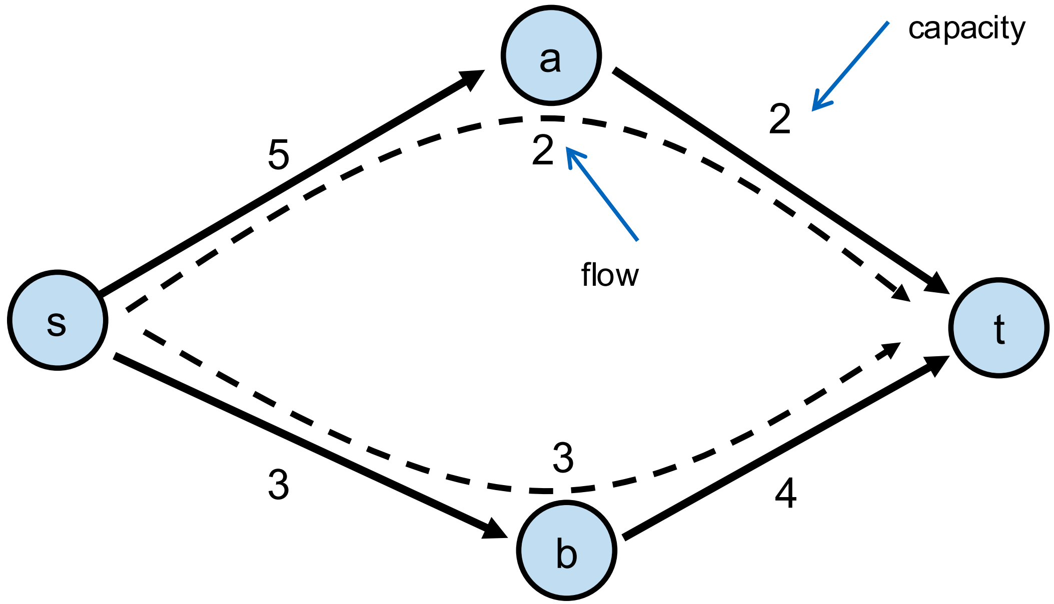
A network $N = (V, E, u, s, t)$ is given by:

- upper capacities for edges u
- special nodes: **source** s und **sink** t

Find the maximum flow that can be sent from the source to the sink:

- Throughput in pipelines
- Data packages on the internet
- Passengers for flights worldwide
- Throughput of a power network

Example



The maximal flow from s to t is 5.

Maximum flow LP formulation

max $\quad x_{sa} + x_{sb}$

s.t. $\quad x_{sa} = x_{at}$

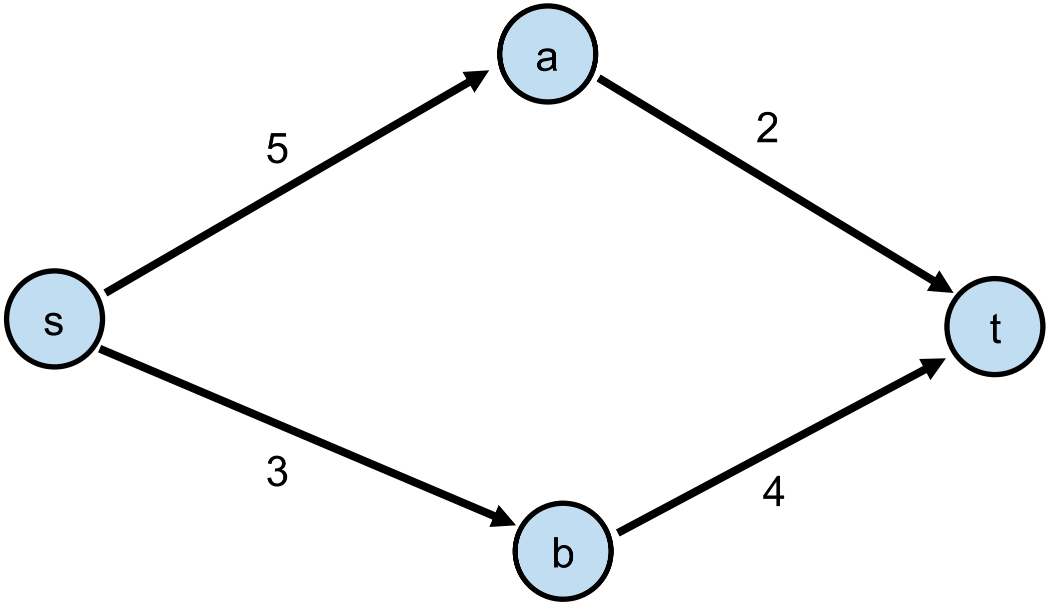
$x_{sb} = x_{bt}$

$0 \leq x_{sa} \leq 5$

$0 \leq x_{sb} \leq 3$

$0 \leq x_{at} \leq 2$

$0 \leq x_{bt} \leq 4$



General maximum flow LP formulation

max	$\sum_{j \in V} x_{sj}$	
s.t.	$\sum_{j \in V} x_{ij} - \sum_{j \in V} x_{ji} = 0$	for each $i \in V, j \neq s, t$
	$x_{ij} \geq 0$	for each $(i, j) \in E$
	$x_{ij} \leq u_{ij}$	for each $(i, j) \in E$

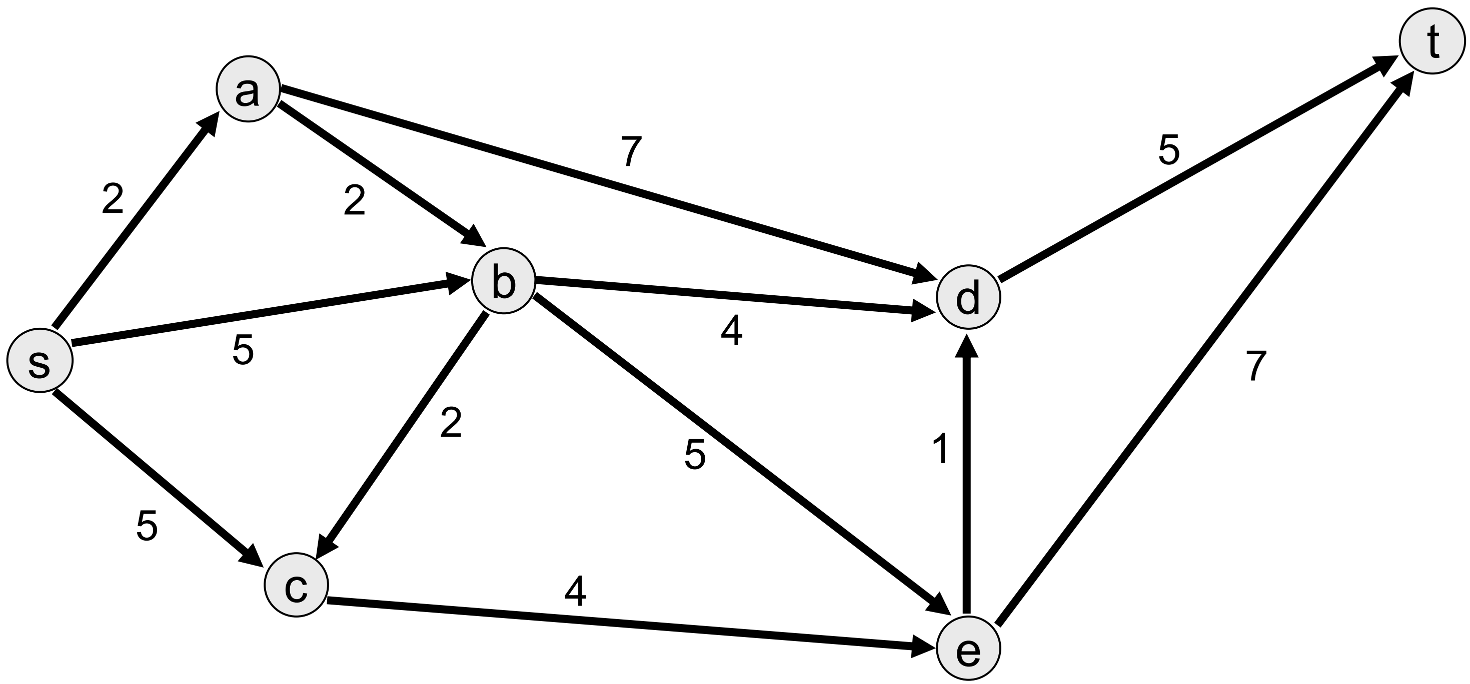
Ford-Fulkerson algorithm

Start with a flow of 0 on all edges.

Repeat:

- Find an (arbitrary) path from s to t (augmented path).
- Find the edge with minimum capacity κ in the path and send flow of value κ along the path.
- Replace the network with the **residual network**: replace each edge e with positive flow $f(e)$ with:
  - **Forward edge** e with capacity $u_e - f(e)$
  - **Backward edge** e' with capacity $f(e)$.
- Until there is no more path with positive capacity from s to t

Example

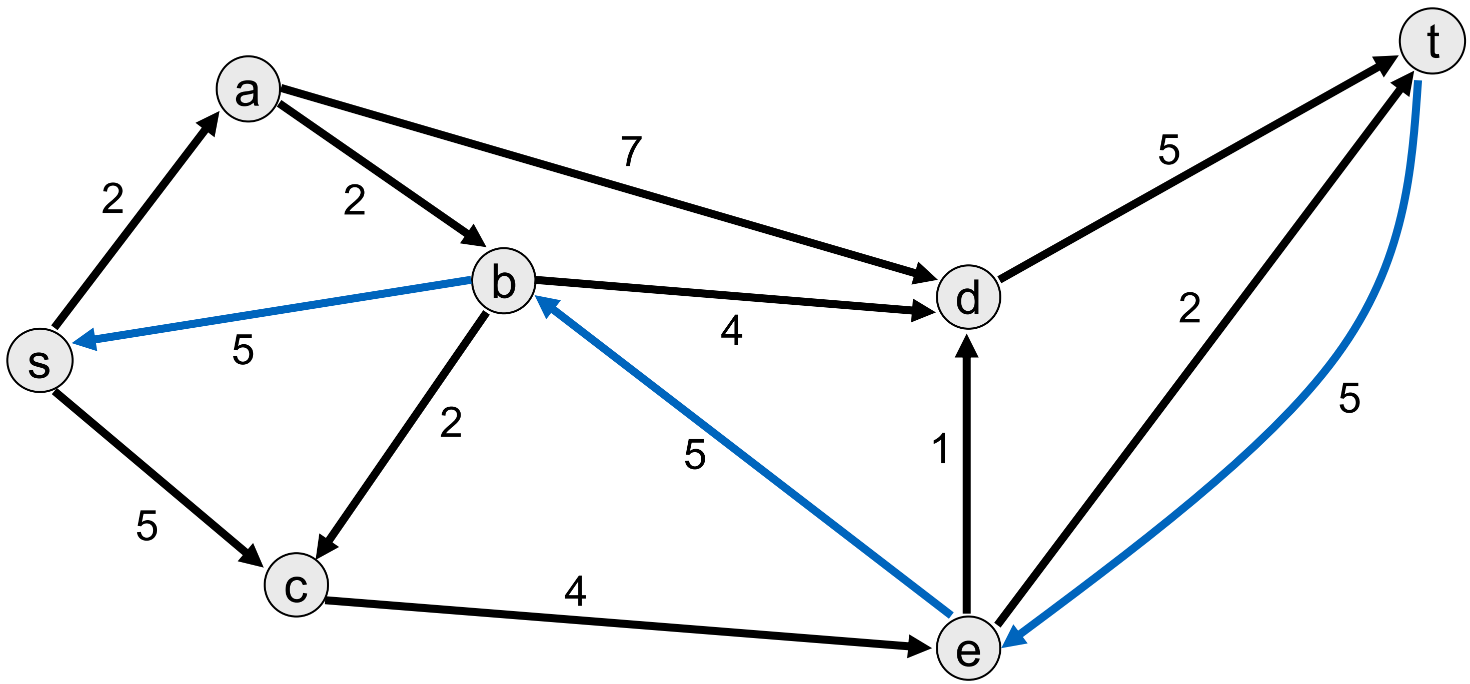


Original network

Choose any path from s to t.

→ Path $s - b - e - t$ with minimum capacity 5

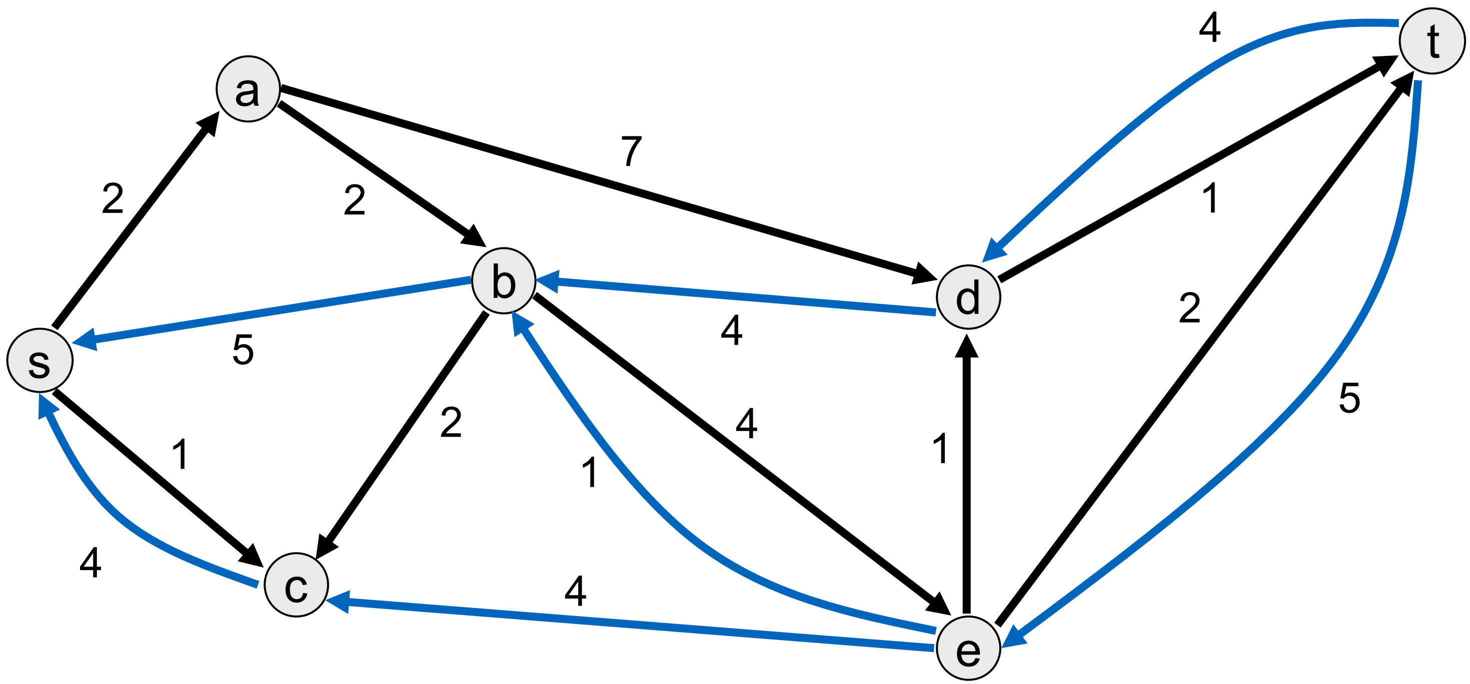
Example



Create the residual network

Find a new $s - t$ path
→ path s – c – e – b – d – t with minimum capacity 4

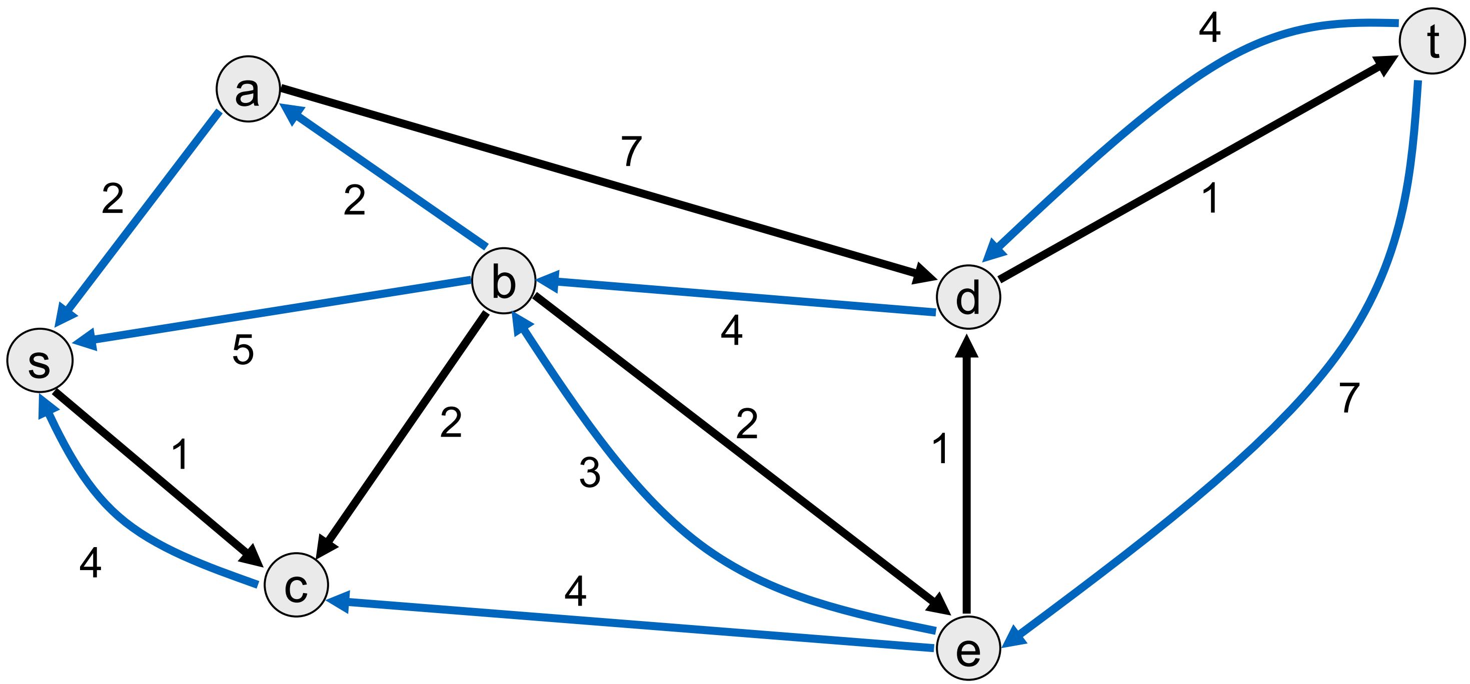
Example



Create the residual network

Find a new $s - t$ path
→ path s – a – b – e – t with minimum capacity 2

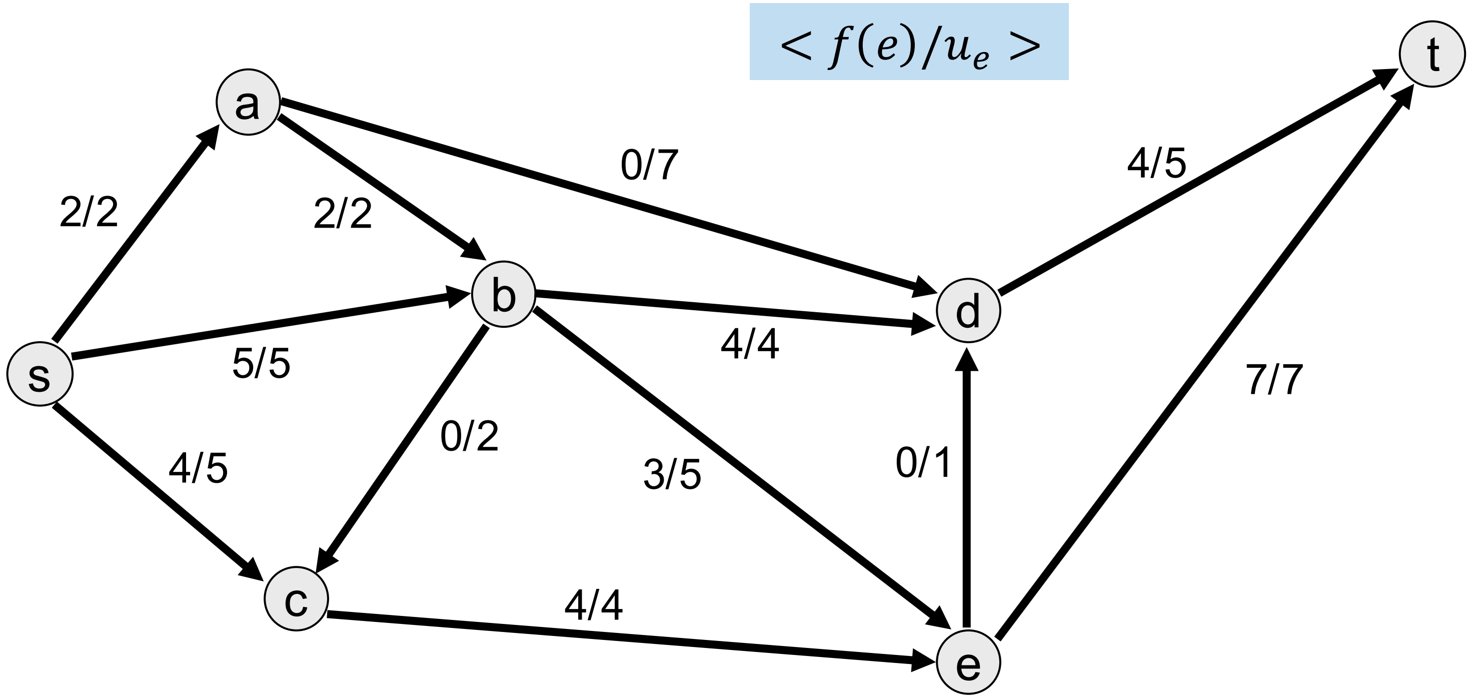
Example



There are no other paths from s to t.
The current flow is optimal.

The remaining forward edges contain residual capacity.

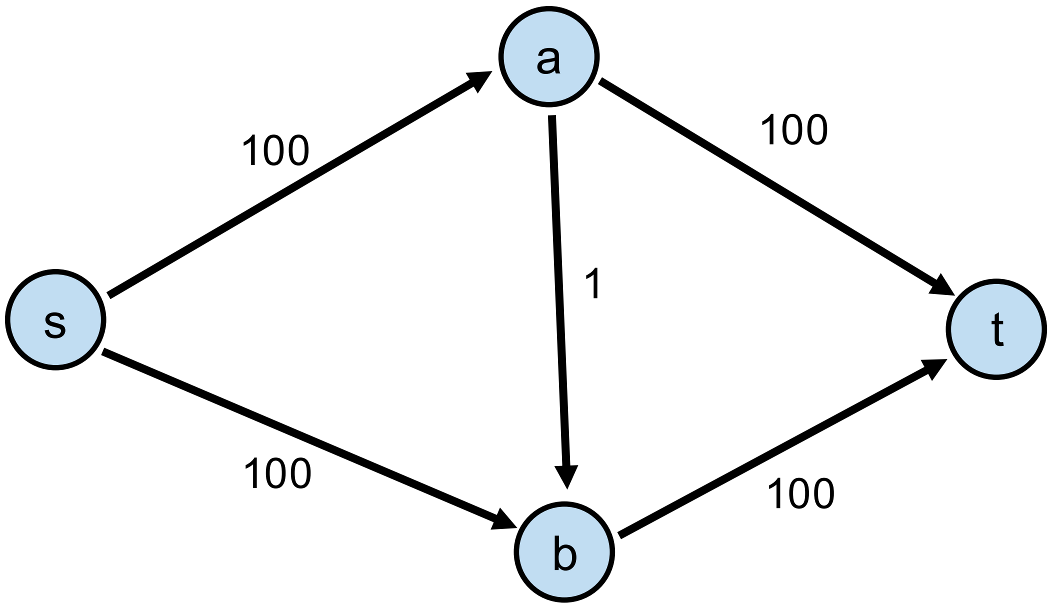
Maximum flow



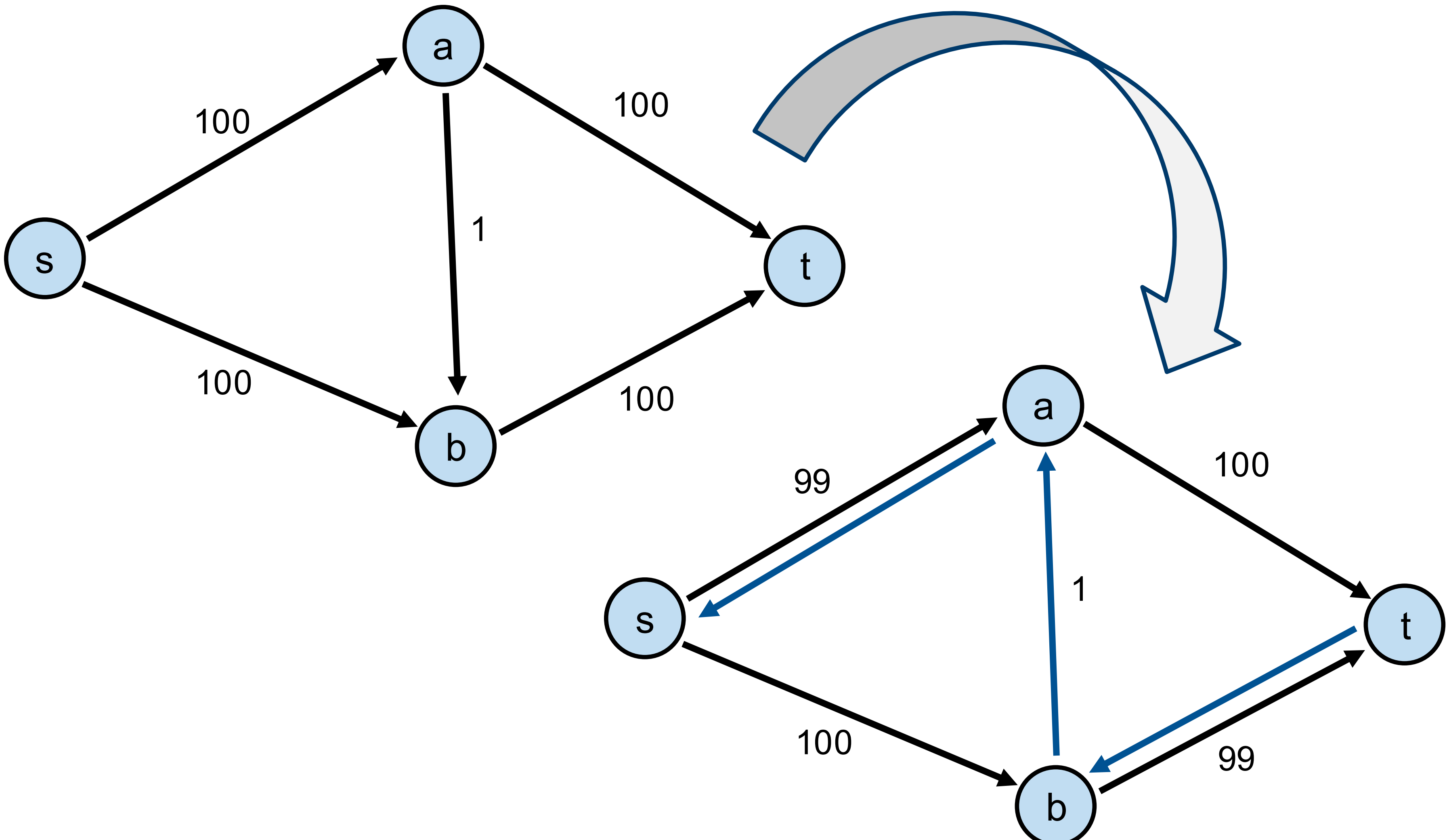
The flow can be read from the backward edges.

→ Maximal flow is 11

Number of steps in Ford-Fulkerson



Number of steps in Ford-Fulkerson



Characteristics

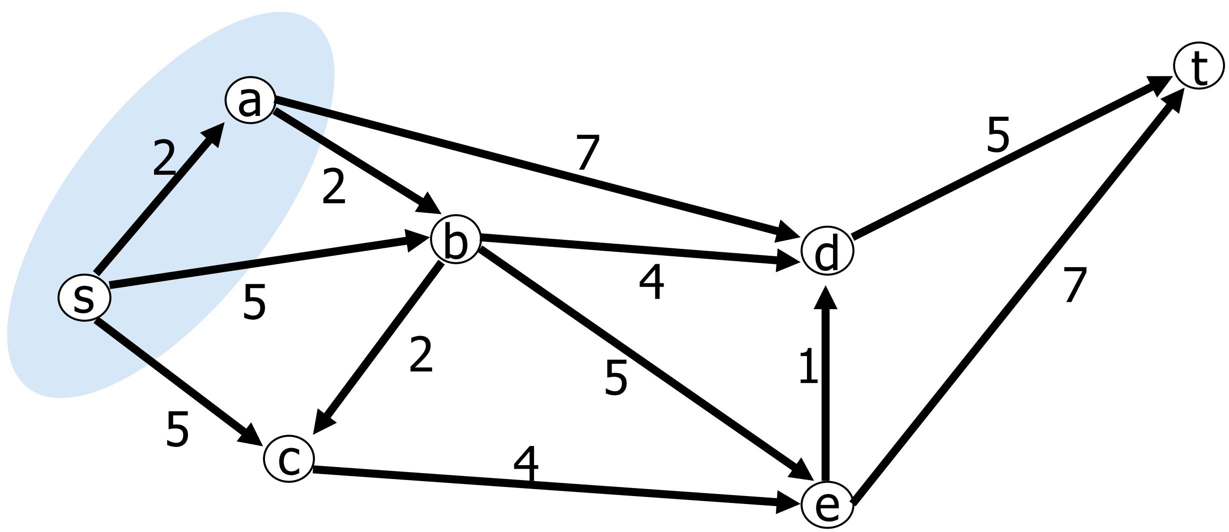
If all capacities of the given network are non-negative integers, the Ford-Fulkerson algorithm stops after finitely many steps and yields a maximum s-t-flow, which is also integer.

- $O(|E| \cdot U)$, with $|E|$ = number of edges and U = maximum capacity of an edge
- So, the algorithm is only *pseudopolynomial* (depends not only on the number of nodes / edges, but also on the size of the numerical parameters).

If we always choose a shortest augmenting path to increase the flow in each step (e.g., using breadth-first search), we obtain the Edmonds-Karp algorithm, which always computes a maximum s-t flow in $O(V \cdot E^2)$.

Cuts and flows

- An **s-t cut** is a partition of nodes $S = [X, V\backslash X]$ such that $s \in X \subset V$ und $t \in V\backslash X$.
- After removing the edges from X to $V\backslash X$, no path exists from s to t.
- **Capacity** of s-t cut: $cap(S) = \sum_{(i,j):i\in X, j\in V\backslash X} u_{ij}$
- **Flow value** of s-t cut: $val(S) = \sum_{(i,j):i\in X, j\in V\backslash X} f(i,j) - \sum_{(j,i):i\in X, j\in V\backslash X} f(j,i)$



$S = [\{a, s\}, \{c, b, d, e, t\}]$ is an s-t-cut with
$\mathrm{cap(S)} = 7 + 2 + 5 + 5 = 19$

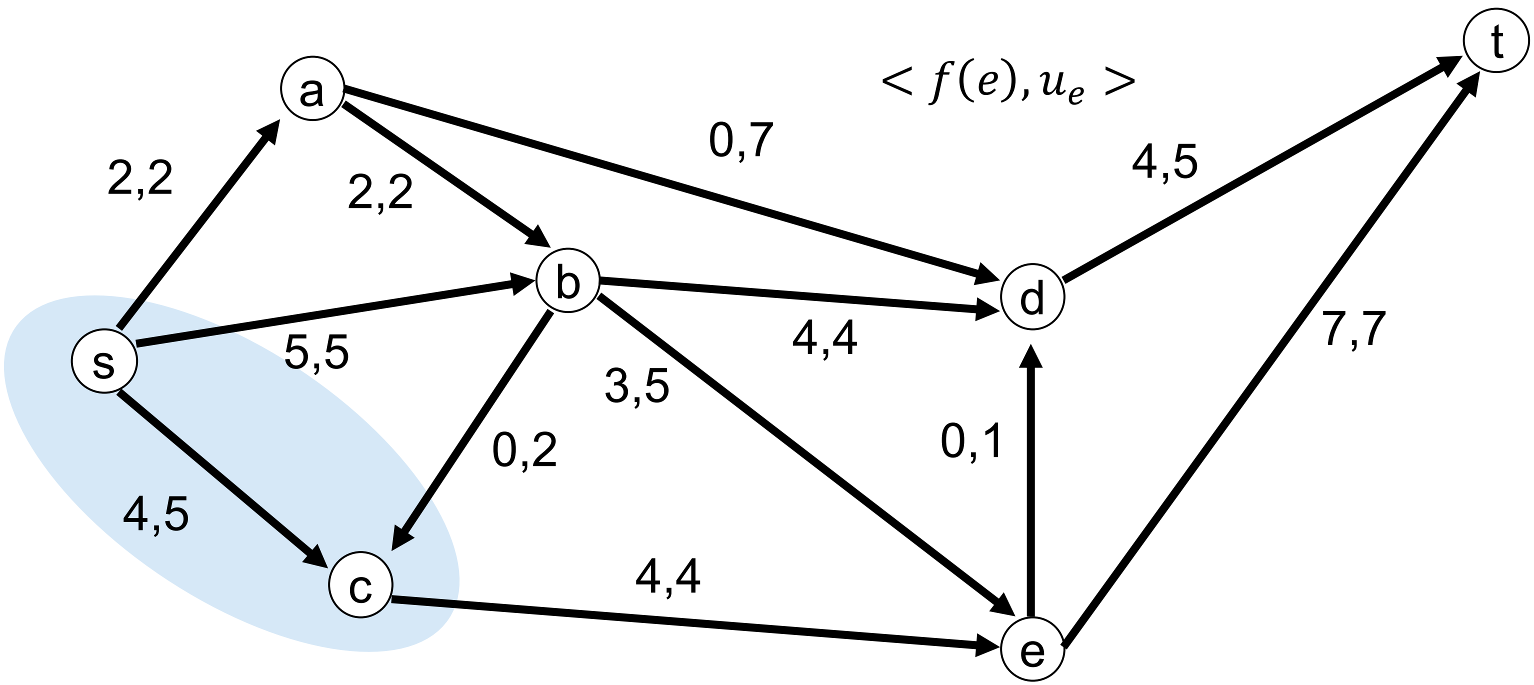
Theorem: Maximum flow - minimum cut

If we consider an s-t cut, each path from s to t contains (at least) one edge of the cut.

The maximum amount of flow that can be sent from s to t cannot exceed the sum of the capacities of a cut.

Therefore, the maximum flow from s to t also corresponds to the minimum cut of all possible s-t cuts.

Example



$S = [\{s, c\}, \{a, b, d, e, t\}]$ is an s-t-cut with $\text{cap}(S) = 11$
and $val(S) = 2 + 5 - 0 + 4 = 11$, where f is maximum.

Maximum flow formulation

max $\quad x_{sa} + x_{sb}$ (dual variables)

s.t. $\quad x_{sa} = x_{ab} + x_{at} \ (p_a)$

$x_{sb} + x_{ab} = x_{bt} \ (p_b)$

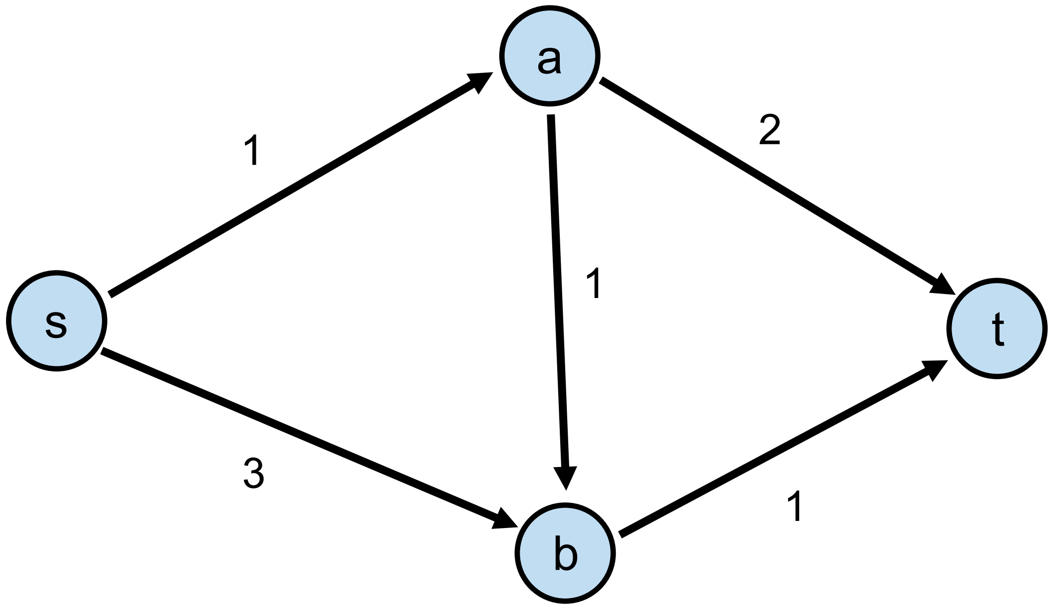
$0 \leq x_{sa} \leq 1 \ (y_{sa})$

$0 \leq x_{sb} \leq 3 \ (y_{sb})$

$0 \leq x_{ab} \leq 1 \ (y_{ab})$

$0 \leq x_{at} \leq 2 \ (y_{at})$

$0 \leq x_{bt} \leq 1 \ (y_{bt})$



Minimum cut formulation (dual of max flow)

min $\quad y_{sa} + 3y_{sb} + y_{ab} + 2y_{at} + y_{bt}$

s.t. $\quad y_{sa} + p_a \geq 1$

$y_{sb} + p_b \geq 1$

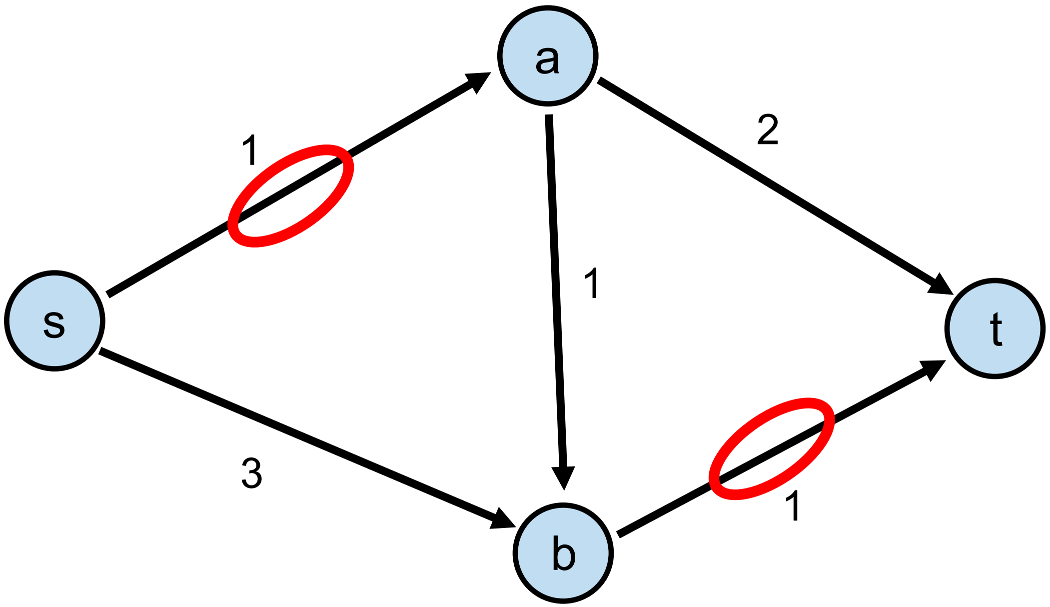
$y_{ab} - p_a + p_b \geq 0$

$y_{at} - p_a \geq 0$

$y_{bt} - p_b \geq 0$

$y_{sa}, y_{sb}, y_{ab}, y_{at}, y_{bt} \geq 0$

$p_a, p_b \in \mathbb{R}$



The forward edges (s,a) and (b,t) are a cut because without these edges there can be no flow from s to t. It is a minimal cut.

Simple and difficult extensions

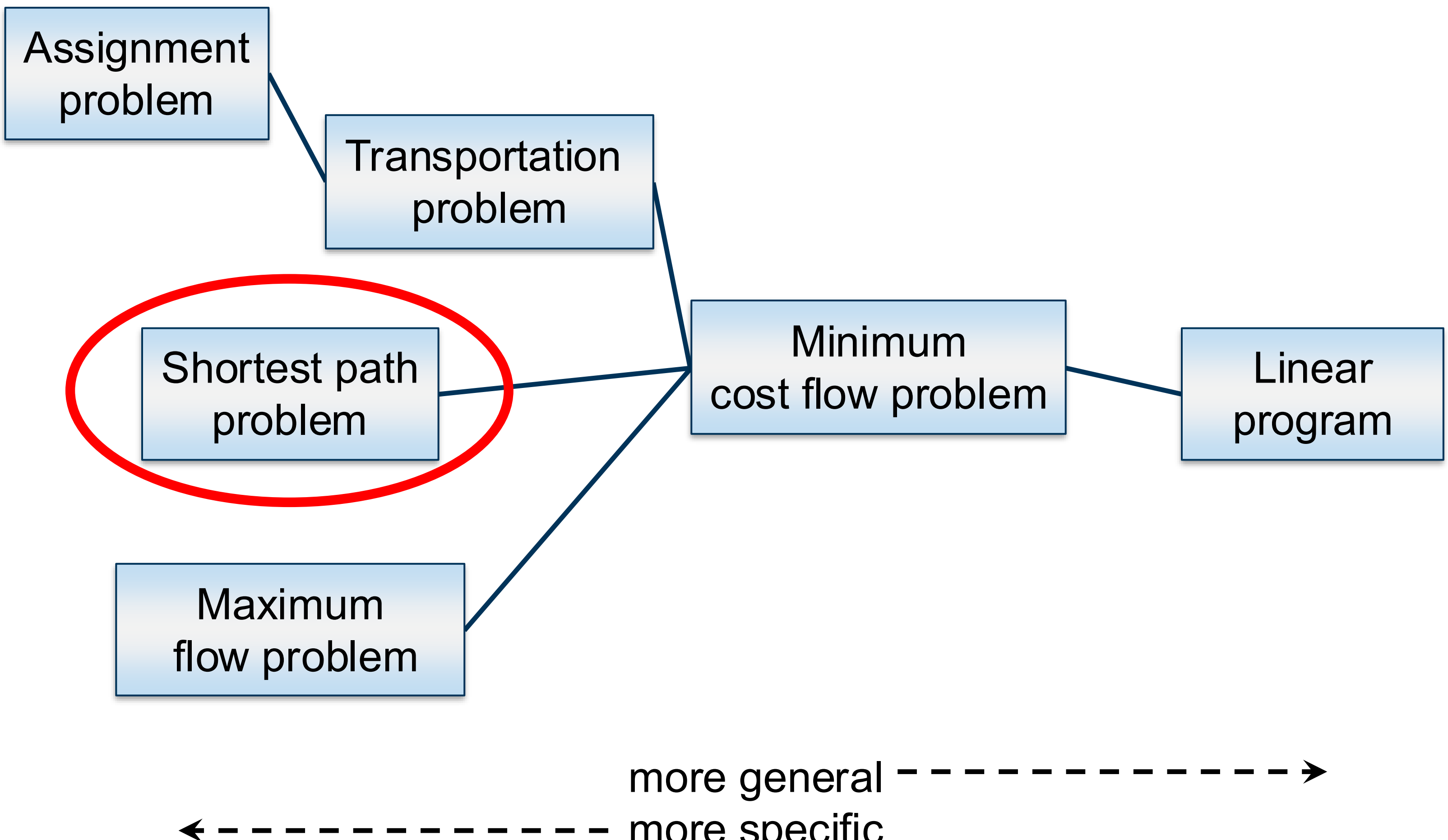
Multiple sources and sinks exist:

- In some cases, there are multiple sources and sinks and we want to determine the total maximum flow from all sources to all sinks.
- We can insert an artificial super source and sink for this purpose.

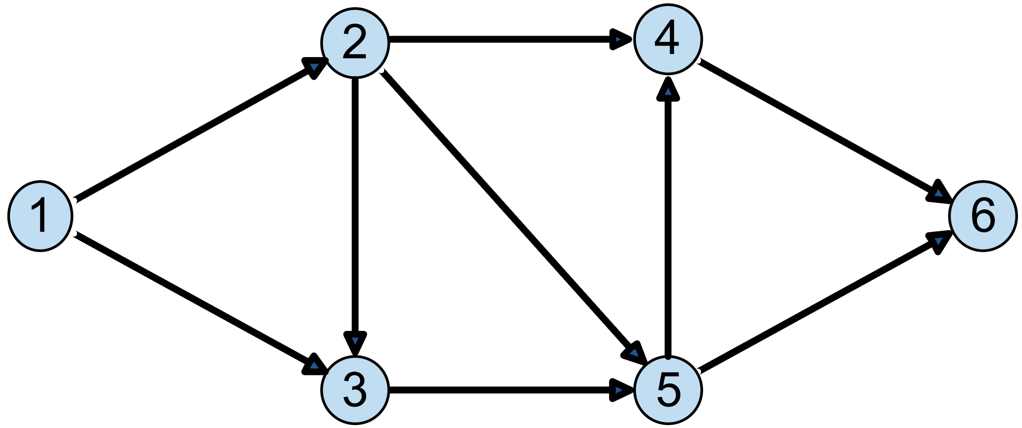
Multiple commodities are sent across the network (multicommodity flow problems):

- The problem becomes $\boldsymbol{NP}$-complete for integer flows.
- But there are approximation algorithms with good approximation guarantees.
- For fractional flows, a solution can be found in polynomial time.

Network flow problems with efficient solutions



Network



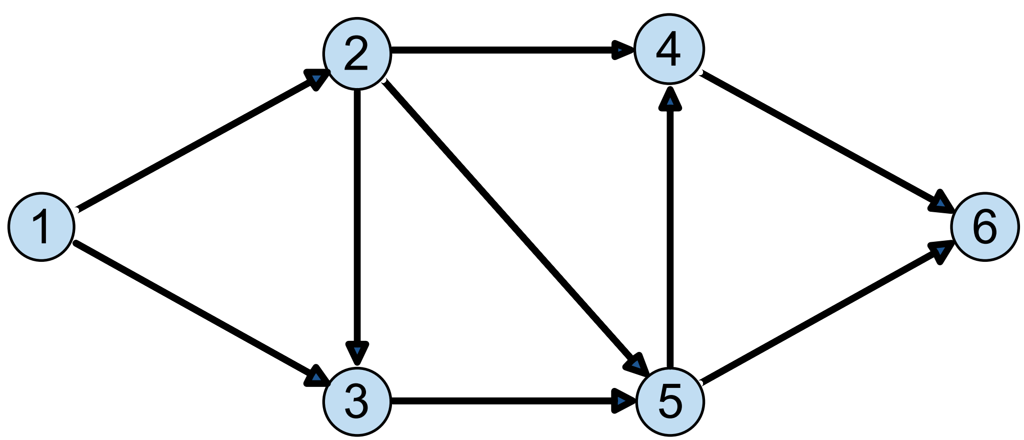
x_{12}	x_{13}	x_{23}	x_{24}	x_{25}	x_{35}	x_{46}	x_{54}	x_{56}		
1	1	0	0	0	0	0	0	0	=	1
-1	0	1	1	1	0	0	0	0	=	0
0	-1	-1	0	0	1	0	0	0	=	0
0	0	0	-1	0	0	1	-1	0	=	0
0	0	0	0	-1	-1	0	1	1	=	0
0	0	0	0	0	0	-1	0	-1	=	-1

Shortest path as MCFP

min $\quad \sum_{(i,j)\in E} c_{ij}x_{ij}$

s.t. $\quad \sum_{j\in V} x_{ij} - \sum_{j\in V} x_{ji} = \begin{cases} 1, & if\ i\ start\ node \\ -1, & if\ i\ end\ node \\ 0, & otherwise \end{cases}$

$x_{ij} \geq 0$ for each $(i,j) \in E$



Summary: Network flow problems

Shortest path

- Dijkstra's algorithm or Bellman-Ford

Allocation problem

- Hungarian method

Transportation problem

- Network simplex

Minimal cost flow (MCF)

- Cycle Cancelling, Minimum Mean Cost Cycles, Network simplex

Maximal flow problems (MFP)

- Ford-Fulkerson, Edmonds-Karp
- Development of faster algorithms until today (Goldberg, Rao, 1997)